

# **Self-Commissioning Drive for PMSM using CCS-MPC with prognostics using Artificial Intelligence**

Zeroth, First & Second Review – Final Year Project

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# 1. Broad Area of Project Work

- **Power Electronics and Electric Drives** – control of Permanent Magnet Synchronous Motors (PMSM)
- **Model Predictive Control (MPC)** based current control schemes – specifically **Continuous Control Set Model Predictive Control (CCS-MPC)**
- **Self-commissioning / auto-tuning drive systems** – automatic estimation of motor parameters without manual intervention
- **Target application domains:** Electric Vehicle traction drives, industrial servo drives, robotics actuators
- Intersection of *control theory*, *power electronics*, and *embedded systems*

## 2. Motivation – Why Does This Problem Matter?

- Standard PI-based FOC relies on linear controllers that struggle to maintain optimal dynamic response, decoupling, and strict constraint handling across the highly non-linear, multi-variable operation.
- Conventional PMSM drives rely heavily on cascaded PI controllers – gains must be **manually tuned** for every individual motor's electrical and mechanical traits
- Manual commissioning is **time-consuming**, requires skilled personnel, and increases factory/EV line downtime
- CCS-MPC gives superior dynamic response, but its inner current prediction model is **highly sensitive** to accurate motor parameters ( $R_s$ ,  $L_d$ ,  $L_q$ ,  $\lambda_f$ )
- A drive that can **self-identify its own parameters** can auto-tune both its outer speed loop (using  $J$ ,  $B$ ) and inner MPC loop, enabling true **plug-and-play** operation
- Strong industry pull from EV and automation sectors for **adaptive, parameter-robust** drive electronics that also flag their own incipient inverter faults

### 3. The Questions We Must Answer

- ❶ **Parameter Estimation:** How can the electrical ( $R_s, L_d, L_q, \lambda_f$ ) and mechanical ( $J, B$ ) parameters of a PMSM be accurately estimated at standstill/low-speed to eliminate manual tuning?
- ❷ **MPC Integration:** How can these identified parameters be directly embedded into a CCS-MPC current predictor (inner loop) and PI speed controller (outer loop) to achieve plug-and-play operation?
- ❸ **Control Robustness:** What is the sensitivity of the CCS-MPC prediction accuracy and closed-loop performance to parameter estimation errors, and how can robust tracking be maintained?
- ❹ **Early Fault Detection:** Can an AI-based prognostic module reliably detect incipient inverter switch-level faults early within a Hardware-in-the-Loop (HIL) environment by reusing the existing current and voltage sensing infrastructure?

## 4. Literature Review (1/3)

| Paper Title                                                                                             | Authors               | Method Adopted                                                                                                 |
|---------------------------------------------------------------------------------------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------|
| (Base Paper) On Continuous-Set Model Predictive Control of Permanent Magnet Synchronous Machines (2022) | I. Hammoud, et al.    | Realizes a real-time CCS-MPC current controller by solving a constrained optimization problem online.          |
| FPGA Implementation of AI-Based Inverter IGBT Open Circuit Fault Diagnosis (2022)                       | N. Rajeswaran, et al. | Implements an AI-based condition monitoring system directly onto an FPGA using neuro-genetic algorithms.       |
| Advanced Deep Learning Approaches for Fault Detection in Inverter-Driven PMSMs (2024)                   | A. Bacha, et al.      | Integrates transformer architectures with Physics-Informed Neural Networks (PINNs) and sensor fusion.          |
| Continuous Control Set Predictive Control with Affine Registration Technique (2024)                     | W. Zhao, et al.       | Proposes a CCS-MPC framework combined with an affine registration technique to mitigate parameter sensitivity. |

## 4. Literature Review (2/3)

| Paper Title                                                                                 | Authors              | Method Adopted                                                                                                  |
|---------------------------------------------------------------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------|
| A Comprehensive Review of Parameter Estimation Techniques for PMSM (2020)                   | M. S. Rafaq, J. Jung | Reviews offline/online parameter estimation techniques to identify methods with fast, accurate convergence.     |
| Global Identification of Electrical and Mechanical Parameters in PMSM Drive (2018)          | Z. Liu, et al.       | Utilizes a dynamic self-learning Particle Swarm Optimization (PSO) to map and estimate parameters.              |
| Real-Time Feasibility of Data-Driven Predictive Control for Synchronous Motor Drives (2023) | P. Carlet, et al.    | Introduces a data-enabled predictive control algorithm to reduce computational complexity for embedded systems. |
| Weight-Adaptable Disturbance Observer for CCS-MPC of NPC-3L-Fed PMSMS (2025)                | Z. Liang, et al.     | Implements a cascaded CCS-MPC strategy with an on-line weight-adaptable disturbance observer.                   |

## 4. Literature Review (3/3)

| Paper Title                                                                                      | Authors          | Method Adopted                                                                                                    |
|--------------------------------------------------------------------------------------------------|------------------|-------------------------------------------------------------------------------------------------------------------|
| An open-circuit fault diagnosis method for PMSM drive system based on multi-branch 1D-CNN (2024) | Y. Zhang, et al. | Utilizes a 1D-CNN to directly extract fault features from raw 3-phase current signals without manual engineering. |
| Mechanism-Based Fault Diagnosis Deep Learning Method for PMSM (2024)                             | L. Li, et al.    | Combines Continuous Wavelet Transform (CWT) with a CNN model to classify specific faults physically.              |

### Research Gap

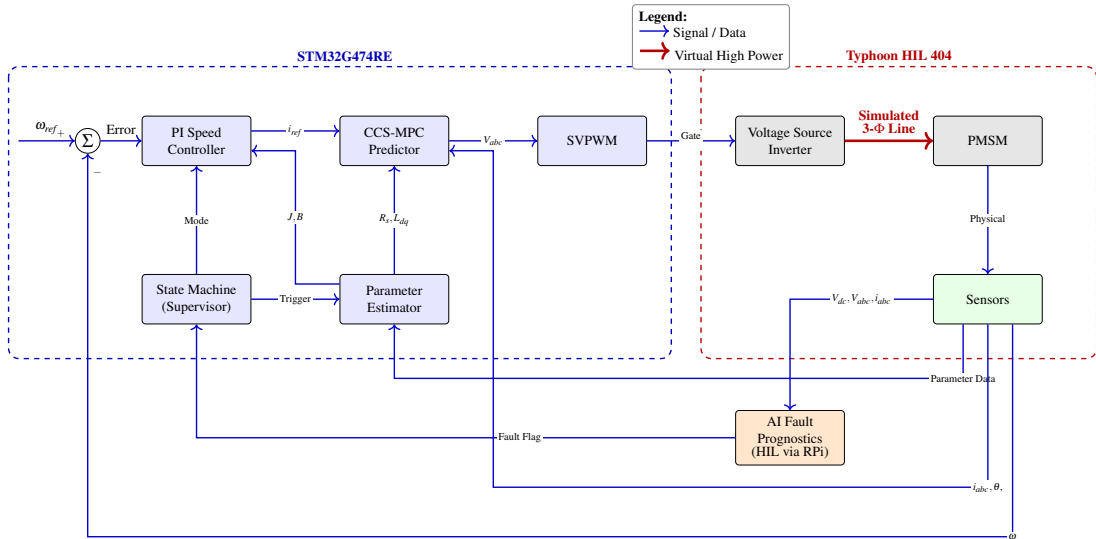
While isolated solutions exist for automated parameter identification, MPC robustness, and AI-based fault diagnostics, there is a lack of a unified framework. Most existing work corrects parameters *during operation* rather than fully self-commissioning prior to standard running, and rarely integrates the drive's data layer simultaneously for **Dual-Loop Auto-tuning** and **AI-based inverter fault prognostics**.

## 5. Our Solution – Specific Objectives

- **Objective 1 (Self-Commissioning Simulation):** Design and simulate a self-commissioning CCS-MPC PMSM drive – standstill DC/AC pulse tests for electrical parameters, and a brief low-speed test for mechanical parameters ( $J, B$ ). Feed  $J, B$  into an auto-tuned PI speed loop, and continuously update the CCS-MPC inner state-space matrices.
- **Objective 2 (MIL & SIL Verification):** Verify the control algorithm through a structured Model-in-the-Loop (MIL) and Software-in-the-Loop (SIL) testing pipeline, utilizing Model-Based Design to auto-generate production C-code for the target microcontroller.
- **Objective 3 (HIL Real-Time Validation):** Validate real-time control performance via Hardware-in-the-Loop (HIL) testing. Execute generated code on the physical STM32G474RE while interfacing with a real-time plant emulator simulating the PMSM and inverter power stage.
- **Objective 4 (AI Prognostics Deployment):** Integrate an AI-based fault prognostic module deployed on a Raspberry Pi. Inject representative inverter switch-level faults within the HIL setup to evaluate real-time signature analysis and anomaly detection.



## 6. Proposed System – The HIL Architecture



## 7. Methodology – The Simulation-to-HIL Pipeline

- **O1 – Simulation & Estimation Setup:** Model PMSM in MATLAB/Simulink. Implement sequences to gather electrical and mechanical parameters dynamically, creating the baseline logic for the control loops.
- **O2 – MIL & SIL Verification:**
  - **Model-in-the-Loop (MIL):** Validate the dual-loop auto-tuning control algorithm against continuous analytical plant models.
  - **Software-in-the-Loop (SIL):** Utilize Embedded Coder to auto-generate C-code from the control logic. Execute this compiled code on the host PC to ensure algorithmic integrity before hardware deployment.
- **O3 – HIL Validation:** Interface the physical **STM32G474RE microcontroller** with the real-time plant emulator. The MCU executes the CCS-MPC and SVPWM natively, passing hardware gate signals to the simulator and receiving analog/digital feedback in real-time.
- **O4 – Prognostics Deployment:** Inject controlled inverter switch-level faults via the HIL software interface. Utilize a **Raspberry Pi** to intercept the real-time analog signatures and execute the AI fault classification model, completing the diagnostic loop.

## 8. Parameter Identification & Auto-Tuning Strategy

### 1. Electrical Parameter Estimation (Standstill)

- **Stator Resistance ( $R_s$ ):** Apply a DC voltage pulse across the phases and measure the resulting steady-state current ( $V/I$ ).
- **Inductances ( $L_d, L_q$ ):** Inject high-frequency voltage pulses or step voltages to analyze the current rate-of-change ( $di/dt$ ) along the  $d$  and  $q$  axes.

### 2. Mechanical Parameter & Flux Estimation Sequence

- **Step 1 - Flux Test ( $\lambda_f$ ):** The drive spins the motor up open-loop, cuts the power, and measures the unpowered Back-EMF voltage coast-down to calculate  $\lambda_f$ .
- **Step 2 - Friction Test ( $B$ ):** Now that it knows  $\lambda_f$ , it spins the motor back up to a constant speed. It measures  $i_q$ , calculates the exact  $T_e$  it is generating, and divides that by the speed to find  $B$ .
- **Step 3 - Inertia Test ( $J$ ):** From that constant speed, it cuts the torque to zero and watches how fast the motor decelerates to find  $J$ .

### 3. PI Speed Loop Auto-Tuning

- The identified parameters dictate the outer speed controller tuning based on a desired closed-loop bandwidth ( $\omega_c$ ).
- The PI gains are calculated.

# 10. Supervisory Control: State Machine, Mode & Trigger

## 1. The State Machine Sequence

- **State 0 (IDLE):** Waiting for start command, switches are OFF.
- **State 1 (STANDSTILL):** Injecting DC/AC pulses to find  $R_s$  and  $L_{dq}$ .
- **State 2 (ALIGNMENT):** Locking the rotor to  $\theta = 0$ .
- **State 3 (OPEN-LOOP SPINUP):** Using Openloop control to reach steady speed.
- **State 4 (COAST-DOWN):** Switches OFF, measuring Back-EMF and deceleration.
- **State 5 (CALCULATE GAINS):** Computing  $K_p, K_i$  and MPC matrices  $A_d, B_d$ .
- **State 6 (NORMAL RUN):** Closing the loops, standard MPC drive operation.

## 2. Mode Signal (To PI Controller)

- **DISABLED:** Outputs zero (during standstill & coast-down).
- **OVERRIDE:** Passes hardcoded open-loop torque limit (during spinup).
- **ACTIVE:** Uses auto-tuned gains.

## 3. Trigger Signal (To Estimator)

- **Capture  $R_s, L_{dq}$ :** Triggered instantly during State 1 pulses.
- **Capture  $\lambda_f, J, B$ :** Triggered at exact moments in States 3 & 4 to sync the math calculation with physical events.

## 11. Expected Outcome – What Success Looks Like

- A working **self-commissioning algorithm** that estimates both electrical ( $R_s, L_{dq}, \lambda_f$ ) and mechanical ( $J, B$ ) parameters within an acceptable error margin
- An **Auto-tuned PI outer loop** and a **CCS-MPC inner current loop** (on STM32G474RE) that dynamically update their internal matrices based purely on self-identified data
- Demonstrated **reduction in commissioning time** compared to conventional manual-tuning workflows
- A comprehensive **Hardware-in-the-Loop testbed** demonstrating adaptive motor control interacting safely with high-fidelity simulated power stages
- An **AI-based prognostic module** capable of early detection of inverter switch-level (SVPWM) faults, fully deployed and validated on a **Raspberry Pi** during HIL fault-injection testing

# 12. Software and Hardware Requirements

## Software Toolchain

- MATLAB / Simulink
- Simscape Electrical (for Plant Modeling)
- Stateflow (supervisory test logic)
- Embedded Coder (C-code generation)
- STM32CubeIDE (for MPC control flash)
- Python / TensorFlow (for AI prognostics on Raspberry Pi)

## Hardware & Emulation

- **Microcontroller (STM32G474RE)** for real-time cascaded control execution
- **Raspberry Pi** (for AI Prognostics deployment)
- Real-time HIL target (Typhoon 404 HIL)
- Interface logic shifters & jumper cabling
- Oscilloscope / Logic Analyzer

## 13. Cost Estimate of the Project

| Item                                  | Purpose                         | Approx. Cost (INR)        |
|---------------------------------------|---------------------------------|---------------------------|
| Microcontroller dev kit (STM32G474RE) | Real-time embedded control      | 3,000 – 5,000             |
| Raspberry Pi kit                      | AI Prognostics Deployment       | 4,000 – 7,000             |
| Interface Components                  | I/O shifting for HIL connection | 1,000 – 2,000             |
| HIL platform usage / lab access       | Real-time plant emulation       | As per lab availability   |
| <b>Estimated Total</b>                |                                 | <b>INR 8,000 – 14,000</b> |

*\*Note: By focusing on HIL validation rather than a physical high-voltage power stage, prototype costs and hardware risk are significantly minimized while preserving control complexity.*

## 14. Time Frame for Execution (8 Months)

| Task                                                                                                                 | M1 | M2 | M3 | M4 | M5 | M6 | M7 | M8 |
|----------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|----|
| Lit. survey & topic finalization                                                                                     | ■  |    |    |    |    |    |    |    |
| Standstill ID & Low-Speed ID logic                                                                                   | ■  | ■  |    |    |    |    |    |    |
| CCS-MPC integration & simulation                                                                                     |    | ■  | ■  |    |    |    |    |    |
| MIL testing & PI Auto-tuning                                                                                         |    |    | ■  | ■  |    |    |    |    |
| SIL testing (C-code generation)                                                                                      |    |    | ■  | ■  | ■  |    |    |    |
| HIL setup & STM32 interfacing                                                                                        |    |    |    | ■  | ■  | ■  |    |    |
| HIL validation of Auto-tuning drive                                                                                  |    |    |    |    | ■  | ■  | ■  |    |
| HIL Inverter Fault Injection sequencing                                                                              |    |    |    |    |    | ■  | ■  | ■  |
| AI Prognostics Training with Raspberry Pi & Integration                                                              |    |    |    |    |    | ■  | ■  | ■  |
| Final HIL System Demo                                                                                                |    |    |    |    |    |    | ■  | ■  |
| Documentation & final review                                                                                         |    |    |    |    |    |    |    | ■  |
| Semester 1: M1–M4 (Algorithm + MIL + SIL)    Semester 2: M5–M8 (HIL Interfacing + AI Prognostics + Final Validation) |    |    |    |    |    |    |    |    |



# 15. References (IEEE Format)

**Base Paper:** I. Hammoud *et al.*, “On Continuous-Set Model Predictive Control of Permanent Magnet Synchronous Machines,” *IEEE Trans. Power Electron.*, vol. 37, no. 9, Sept. 2022.



M. S. Rafaq and J. Jung, “A Comprehensive Review of State-of-the-Art Parameter Estimation Techniques for PMSM,” *IEEE Trans. Ind. Inform.*, vol. 16, no. 7, 2020.



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P. Carlet *et al.*, “Real-Time Feasibility of Data-Driven Predictive Control for Synchronous Motor Drives,” *IEEE Trans. Power Electron.*, vol. 38, no. 2, 2023.



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Y. Zhang *et al.*, “An open-circuit fault diagnosis method for PMSM drive system based on multi-branch 1D-CNN,” *J. Phys.: Conf. Ser.*, 2024.



L. Li *et al.*, “Mechanism-Based Fault Diagnosis Deep Learning Method for PMSM,” *Sensors*, vol. 24, no. 19, 2024.



M. Gierczyński and L. M. Grzesiak, “Comparative Analysis of the Steady-State Model Including Non-Linear Flux Linkage Surfaces and the Simplified Linearized Model when Applied to a Highly-Saturated Permanent Magnet Synchronous Machine—Evaluation Based on the Example of the BMW i3 Traction Motor,” *Energies*, vol. 14, no. 9, art. 2343, 2021.

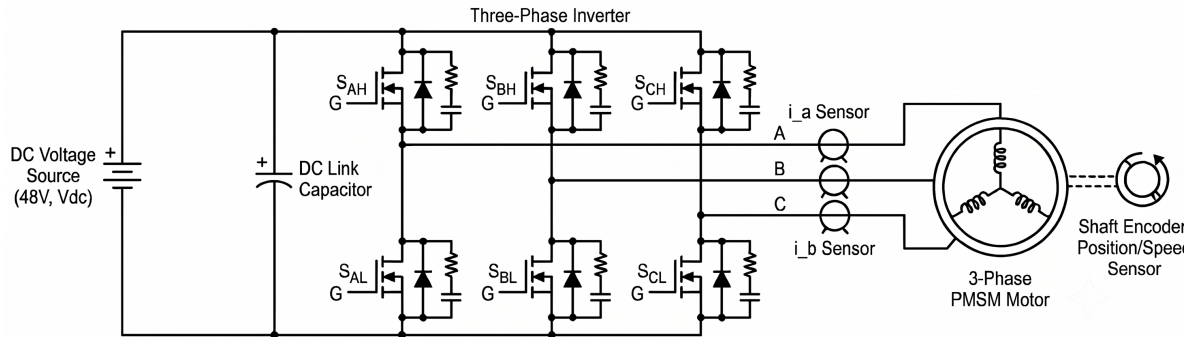
# First Review

Progress & Architecture Analysis

## 16. The Industry Problem & Our Solution

- **The Industry Standard:** Permanent Magnet Synchronous Motors (PMSMs) dominate EV traction and industrial automation due to high efficiency. However, their drives rely almost entirely on Field-Oriented Control (FOC) governed by manually tuned PI controllers.
- **The Bottleneck:** Whenever a motor is swapped on an assembly line, technicians must halt production to manually re-tune the  $K_p$  and  $K_i$  gains to match the new motor's specific electrical and mechanical traits.
- **Why CCS-MPC for PMSM:** PMSMs exhibit highly coupled, non-linear d-q axis dynamics (especially at high speeds). CCS-MPC explicitly handles these cross-couplings and multi-variable constraints analytically, enabling superior transient response without cascaded PI tuning.
- **Our Proposed Solution:**
  - ① Replace the rigid FOC current loops with a **Continuous Control Set Model Predictive Control (CCS-MPC)** algorithm that analytically calculates optimal voltages.
  - ② Introduce an **AutoTuner State Machine** to autonomously execute pre-operational self-commissioning (extracting  $R_s, L_d, L_q, \Psi_{PM}, J, B$ ) and directly feed them to the MPC, enabling true plug-and-play capability.

## 16a. Plant Model: Power Stage



# 16b. System Parameters (Plant & Controller)

## PMSM & Inverter Parameters

*Standard Application: BMW i3 Traction Motor (EV) [10]*

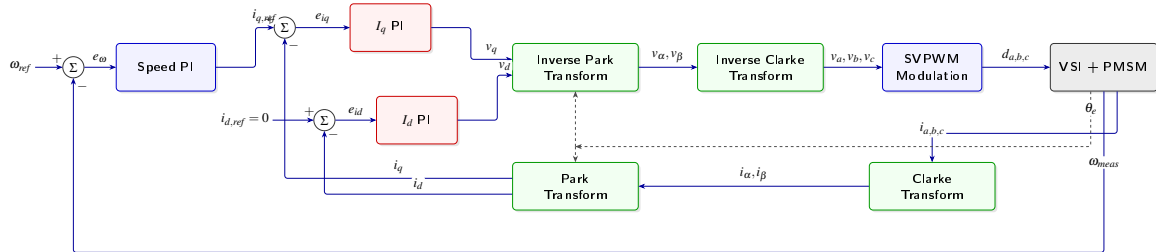
| Parameter              | Symbol      | Value                       |
|------------------------|-------------|-----------------------------|
| Nominal DC Bus Voltage | $V_{dc}$    | 275.7 V                     |
| MOSFET On-Resistance   | $R_{on}$    | 1.0 m $\Omega$              |
| Switching Frequency    | $f_{sw}$    | 10 kHz                      |
| Stator Resistance      | $R_s$       | 5.3 m $\Omega$              |
| d/q-axis Inductance    | $L_d, L_q$  | 71.2 $\mu$ H, 141.3 $\mu$ H |
| PM Flux Linkage        | $\Psi_{PM}$ | 0.0436 Wb                   |
| Pole Pairs             | $p$         | 6                           |
| Rotor Inertia          | $J$         | 0.0867 kg·m <sup>2</sup>    |
| Viscous Friction       | $B_{fric}$  | 0.001 N·m·s/rad             |
| Torque Constant        | $K_t$       | 0.392 N·m/A                 |
| Peak Torque            | $T_{rated}$ | 258.2 N·m                   |

## Control Loop & Simulation

| Parameter             | Symbol                 | Value                    |
|-----------------------|------------------------|--------------------------|
| Control Sample Time   | $T_{s,ctrl}$           | 100 $\mu$ s              |
| Simulation Time Step  | $T_{s,sim}$            | 5 $\mu$ s                |
| Speed PI Gains        | $K_{p,spd}, K_{i,spd}$ | 2.0, 10.0                |
| Current PI Bandwidth  | $\omega_{c,curr}$      | $2\pi \times 1000$ rad/s |
| Max Phase Current     | $I_{peak}$             | 565.7 A                  |
| CCS-MPC Error Weights | $Q_{id}, Q_{iq}$       | 1.0, 1.0                 |
| CCS-MPC Switch Weight | $R_{sw}$               | 0.001                    |
| Max Phase Voltage     | $V_{ph,max}$           | 159.2 V                  |
| Inverter Deadtime     | $T_{dead}$             | 2 $\mu$ s                |

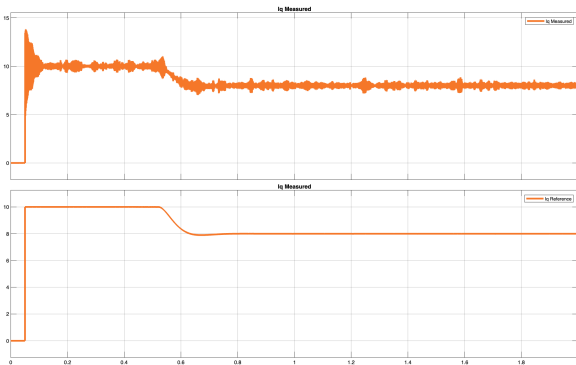
*Note: Parameters verified via Comparative Analysis of Steady-State Models [6].*

## 17. Conventional Field-Oriented Control (FOC) Architecture

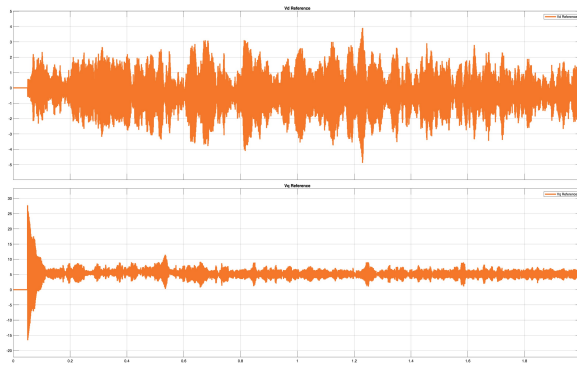


## 17a. FOC Simulation Results – Constant Speed: Current Tracking

### $I_q$ Tracking



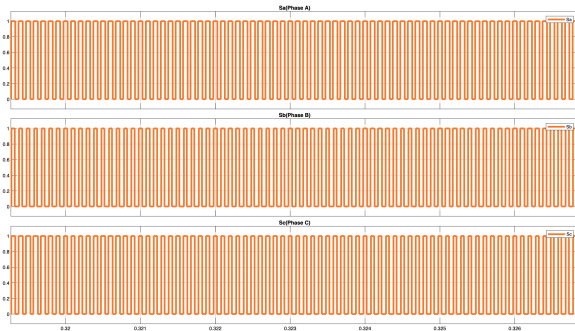
### $V_d, V_q$ Tracking



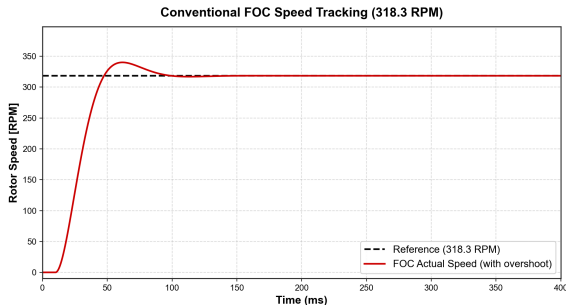
- **Current Tracking Dynamics:** FOC current tracking under constant speed reference shows visible ripple on both axes due to cascaded PI loop delays and cross-axis coupling.

## 17b. FOC Simulation Results – Constant Speed: Switching & Speed

### Switching Pulses ( $S_a, S_b, S_c$ )



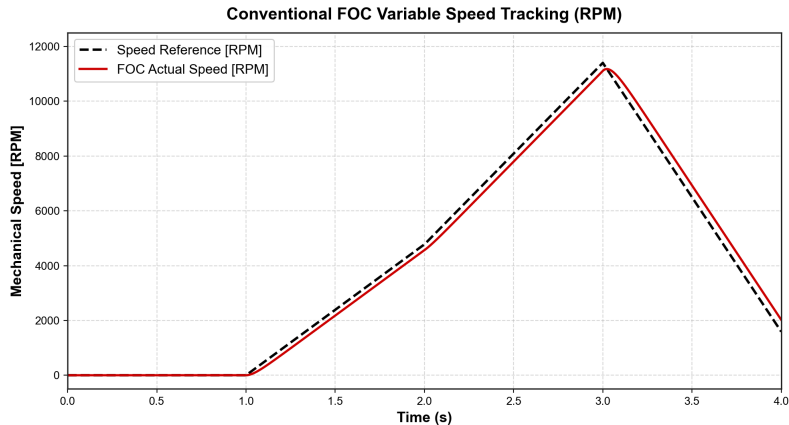
### Speed Tracking



- **Speed Response:** The FOC speed response settles around the 318.3 RPM reference (200 rad/s electrical) after an initial overshoot, with steady switching action from the SVPWM stage.

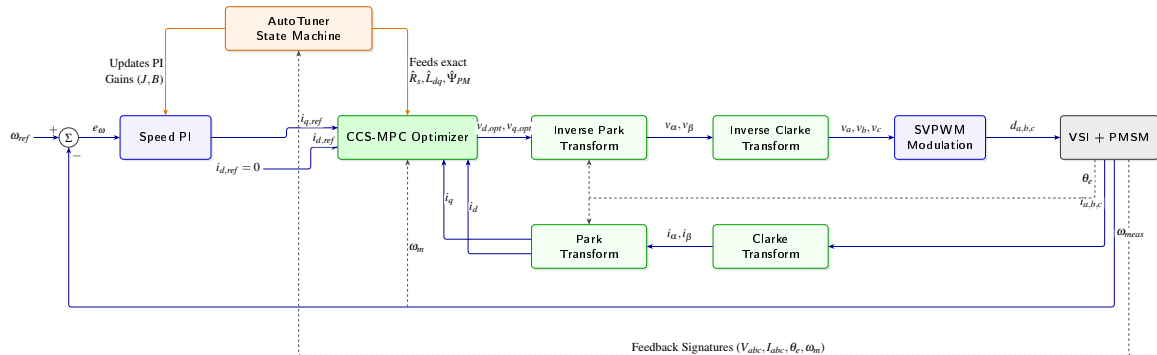


## 17c. FOC Simulation Results – Variable Speed Tracking



- **Tracking Lag:** Under multi-step variable speed profiles (ramp – hold – ramp), the FOC speed loop exhibits visible lag during transitions and increased ripple as speed rises.

## 18. Our Solution: CCS-MPC Architecture



# 19. CCS-MPC: Predictive Model & Cost Function

**Predictive Model (Forward Euler Discretization):**

$$i_d(k+1) = i_d(k) + \frac{T_s}{L_d} (v_d(k) - R_s i_d(k) + \omega_e(k) L_q i_q(k)) \quad (1)$$

$$i_q(k+1) = i_q(k) + \frac{T_s}{L_q} (v_q(k) - R_s i_q(k) - \omega_e(k) (L_d i_d(k) + \Psi_{PM})) \quad (2)$$

**Deriving Optimal Voltages ( $v_{d,opt}$  &  $v_{q,opt}$ ):**

❶ **Define Unconstrained Cost Function:**

$$J_{cost} = (i_{d,ref} - i_d(k+1))^2 + (i_{q,ref} - i_q(k+1))^2 \quad (3)$$

❷ **Analytic Minimization:** Differentiate cost function and equate to zero  $\left( \frac{\partial J_{cost}}{\partial v_{dq}} = 0 \right)$ .

❸ **Extract Closed-Form Solution:** Solve for the exact voltages:

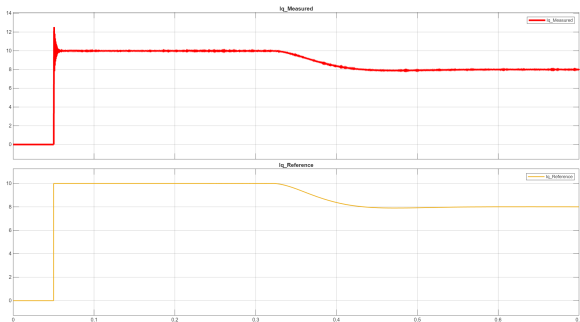
$$v_{d,opt} = \frac{L_d}{T_s} (i_{d,ref} - i_d(k)) + R_s i_d(k) - \omega_e L_q i_q(k) \quad (4)$$

$$v_{q,opt} = \frac{L_q}{T_s} (i_{q,ref} - i_q(k)) + R_s i_q(k) + \omega_e (L_d i_d(k) + \Psi_{PM}) \quad (5)$$

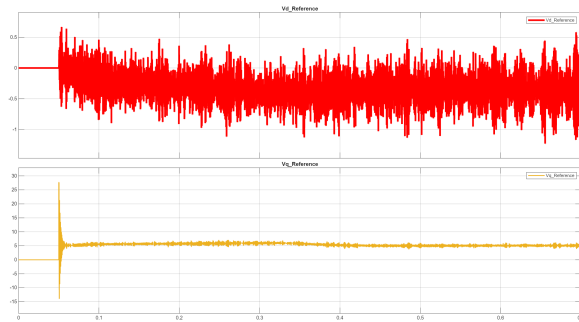
*Note: Equations 1-5 are derived from standard AC machine dynamics (Krause, P.C.) [6]. Vector is clamped to limit ( $V_{dc}/\sqrt{3}$ ) before SVPWM.*

## 19a. CCS-MPC Simulation Results – Constant Speed: Current & Voltage

### $I_q$ Tracking



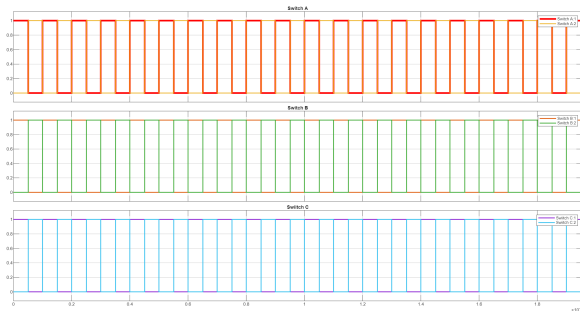
### $V_d, V_q$ Tracking



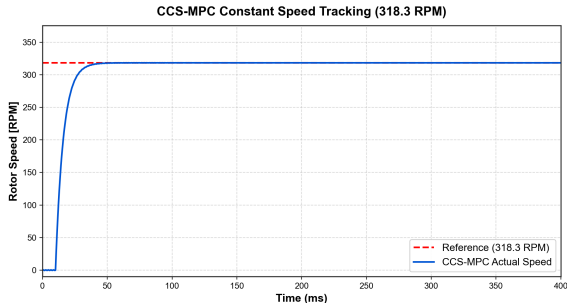
- **Fast Transient Settling:** The deadbeat CCS-MPC current loop settles  $i_q$  to its reference within  $< 2$  ms, with optimal voltages  $v_{d,opt}, v_{q,opt}$  converging immediately.

## 19b. CCS-MPC Simulation Results – Constant Speed: Switching & Speed

### Switching Signals (A, B, C)

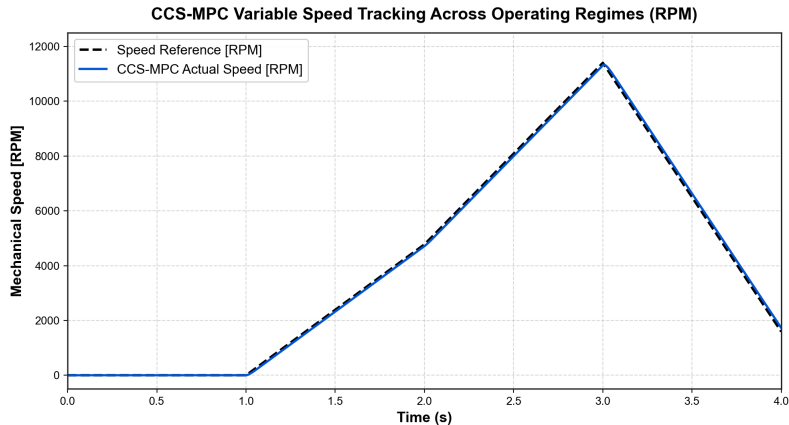


### Speed Tracking



- **Tight Speed Regulation:** The CCS-MPC speed loop reaches 318.3 RPM reference (200 rad/s electrical) quickly and settles into a tight steady-state band with regular 3-phase switching.

## 19c. CCS-MPC Simulation Results – Variable Speed Tracking



- **Trajectory Tracking Accuracy:** Across multi-step variable speed profiles, CCS-MPC tracks each acceleration ramp and hold segment closely with minimal phase lag.

## 20. FOC vs CCS-MPC: Varying Load Comparison

**Summary Table (Base Torque = -250 Nm, Speed = 1273.2 RPM (800 rad/s electrical), Sim Time = 0.6s)**

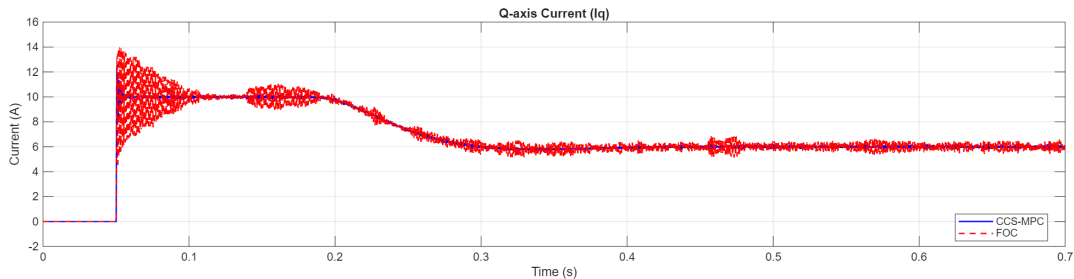
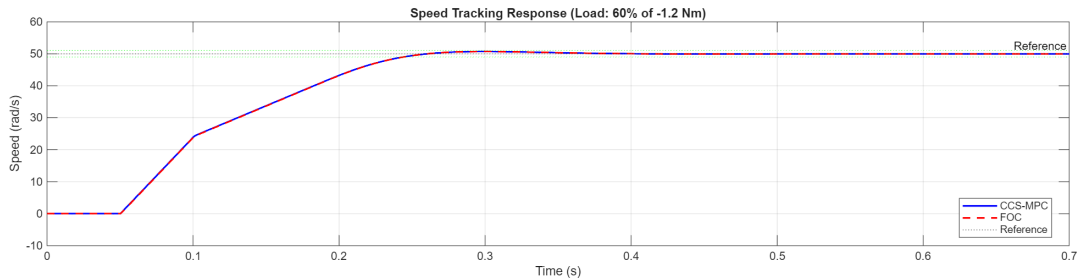
| Load (%)   | Controller     | Max Spd  | Rise (s) | Settl (s) | Over (%)      | SS Err  | Spd Rip        | Iq Rip (A)      | Trq Rip (Nm)   | THD (%)       |
|------------|----------------|----------|----------|-----------|---------------|---------|----------------|-----------------|----------------|---------------|
| <b>100</b> | <b>CCS-MPC</b> | 801.1968 | 0.3599   | 0.4419    | <b>0.1496</b> | 45.0467 | <b>57.8963</b> | <b>89.2013</b>  | <b>10.7042</b> | <b>1.0047</b> |
|            | FOC            | 807.7869 | 0.3548   | 0.4472    | 0.9734        | 47.8207 | 58.6633        | 280.2387        | 33.6286        | 13.3310       |
| <b>80</b>  | <b>CCS-MPC</b> | 801.1710 | 0.3599   | 0.4416    | <b>0.1464</b> | 44.4877 | <b>57.4289</b> | <b>77.1475</b>  | <b>9.2577</b>  | <b>1.8818</b> |
|            | FOC            | 806.7453 | 0.3550   | 0.4450    | 0.8432        | 47.1661 | 58.4969        | 268.9937        | 32.2792        | 28.2914       |
| <b>60</b>  | <b>CCS-MPC</b> | 801.2991 | 0.3598   | 0.4414    | <b>0.1624</b> | 44.4882 | <b>57.4287</b> | <b>94.2956</b>  | <b>11.3155</b> | <b>3.1891</b> |
|            | FOC            | 805.9658 | 0.3553   | 0.4430    | 0.7457        | 46.4910 | 57.9538        | 253.6331        | 30.4360        | 23.2509       |
| <b>50</b>  | <b>CCS-MPC</b> | 801.3479 | 0.3598   | 0.4415    | <b>0.1685</b> | 44.4885 | <b>57.4287</b> | <b>93.4703</b>  | <b>11.2164</b> | <b>3.0972</b> |
|            | FOC            | 805.2953 | 0.3561   | 0.4431    | 0.6619        | 45.8946 | 58.4370        | 265.6205        | 31.8745        | 22.6466       |
| <b>20</b>  | <b>CCS-MPC</b> | 801.0333 | 0.3599   | 0.4415    | <b>0.1292</b> | 44.4890 | <b>57.4289</b> | <b>118.1778</b> | <b>14.1813</b> | <b>0.1741</b> |
|            | FOC            | 804.9702 | 0.3596   | 0.4425    | 0.6213        | 45.0811 | 58.0787        | 237.2207        | 28.4665        | 1.7142        |
| <b>0</b>   | <b>CCS-MPC</b> | 801.2315 | 0.3595   | 0.4410    | <b>0.1539</b> | 44.4875 | <b>57.4277</b> | <b>132.0402</b> | <b>15.8448</b> | <b>0.2859</b> |
|            | FOC            | 805.7548 | 0.3596   | 0.4413    | 0.7193        | 44.2470 | 58.0075        | 231.2172        | 27.7461        | 5.7255        |

### Load Condition Summary

**Heavy / Rated Load Performance (50–100%):** CCS-MPC proves vastly superior, suppressing torque ripple by  $3.14\times$  (10.70 Nm vs 33.63 Nm at 100%) and  $I_q$  ripple by  $3.49\times$  (77.15 A vs 268.99 A at 80%), while maintaining ultra-low THD (1.00% vs 13.33%).

**Light / No-Load Performance (0–20%):** CCS-MPC exhibits tight speed overshoot ( $\leq 0.15\%$  vs up to 0.72%) and superior harmonic suppression (0.17% vs 1.71% THD at 20% load).

## 20a. FOC vs CCS-MPC: Direct Overlay Comparison with 60% load





## 21. Why We Need CCS-MPC (Load Analysis)

- **Vastly Superior Current Tracking:** Across all load conditions, CCS-MPC suppresses  $I_q$  ripple by a massive margin. At 100% load,  $I_q$  ripple is reduced by  $3.14\times$  (89.20 A vs 280.24 A), proving the analytic deadbeat optimizer isolates electrical noise far better than cascaded PI loops.
- **Major Torque Ripple Suppression:** Torque ripple is strictly minimized across the entire torque envelope. At 80% load, CCS-MPC reduces torque ripple by  $3.49\times$  (9.26 Nm vs 32.28 Nm), which directly eliminates acoustic whine and reduces gearbox torsional wear.
- **Superior Speed Regulation & Lower Overshoot:** Across all operating loads, CCS-MPC maintains speed overshoot below 0.17% (1275.3 RPM max), whereas FOC exhibits up to 0.97% overshoot (1285.7 RPM peak).
- **Dramatic THD Reduction:** At 100% load, current THD is reduced by over  $13\times$  (1.00% for CCS-MPC vs 13.33% for FOC), significantly lowering stator harmonic copper losses and thermal rise.
- **Zero Manual Gain Tuning:** Achieving these high-stability current and torque margins requires zero manual PI gain calibration; the controller synthesizes deadbeat voltages directly from identified machine parameters ( $\hat{R}_s, \hat{L}_d, \hat{L}_q, \hat{\Psi}_{PM}$ ).

## 22. Methodology: Electrical Parameter Estimation (Phase 1)

- **Standstill Testing Principle** ( $\omega_e = 0$ ): Rotational Back-EMF is zero, isolating stator impedance ( $R_s, L_d, L_q$ ) without rotor motion dynamics.
- **Stator Resistance ( $\hat{R}_s$ ) Identification via Two-Level DC Injection:**
  - Injects two distinct stationary DC current levels:  $I_{d1} = 0.10I_{max}$  and  $I_{d2} = 0.20I_{max}$ .
  - Evaluates differential resistance to eliminate inverter semiconductor forward voltage drop ( $V_d$  bias):

$$\hat{R}_s = \frac{V_{d2} - V_{d1}}{I_{d2} - I_{d1}} \quad (\text{calculated after 2000 samples at steady state}) \quad (6)$$

- **d-q Inductances ( $\hat{L}_d, \hat{L}_q$ ) Identification via 500 Hz High-Frequency AC Injection:**
  - Injects high-frequency voltage  $v_{dq, inj} = V_{HF} \sin(\omega_{HF} t)$  ( $\omega_{HF} = 2\pi \cdot 500$  rad/s).
  - Extracts fundamental current response via orthogonal Fourier integration over 4000 discrete samples:

$$I_{HF} = \frac{2}{N} \sqrt{(\sum i_{dq} \sin \omega_{HF} t)^2 + (\sum i_{dq} \cos \omega_{HF} t)^2}, \quad Z = \frac{V_{HF}}{I_{HF}} \quad (7)$$

$$\hat{L}_{d,q} = \frac{\sqrt{Z^2 - \hat{R}_s^2}}{\omega_{HF}} \quad (8)$$

## 23. Methodology: Mechanical Parameter & Flux Estimation (Phase 2)

### • Permanent Magnet Flux Linkage ( $\hat{\Psi}_{PM}$ ) via High-Speed Coast-Down:

- Spins motor open-loop past minimum threshold speed  $\omega_{min} = 1273.2$  RPM (800 rad/s electrical).
- Sets stator current commands to zero ( $i_d = 0, i_q = 0$ ) and captures pure Back-EMF:

$$\hat{\Psi}_{PM} = \frac{1}{N} \sum_{k=1}^N \frac{v_q(k)}{\omega_e(k)} \quad (N = 4000 \text{ samples}) \quad (9)$$

### • Viscous Friction Coefficient ( $\hat{B}$ ) via Two-Point Speed Settling:

- Regulates closed-loop mechanical speeds to  $\omega_1 = 573$  RPM and  $\omega_2 = 1146$  RPM (60 rad/s and 120 rad/s) with strict 1.5 s settling timers.
- Evaluates differential electromagnetic torque to eliminate load offsets:

$$\hat{B} = \frac{|\bar{T}_{e,2} - \bar{T}_{e,1}|}{|\bar{\omega}_{m,2} - \bar{\omega}_{m,1}|} \quad \text{where } \bar{T}_e = \frac{3}{2} p \hat{\Psi}_{PM} \bar{I}_{q,cmd} \quad (10)$$

### • Rotor Inertia ( $\hat{J}$ ) via Active Braking Step:

- Injects dynamic braking current step ( $i_{q,ref} = -I_{spin}$ ) and measures deceleration rate  $\frac{d\omega}{dt}$ :

$$\hat{J} = \frac{|T_{e,brake} - \hat{B} \bar{\omega}_{avg}|}{|d\omega_m/dt|} \quad \text{evaluated over 2000 samples} \quad (11)$$

## 24. Self-Commissioning: Parameter Identification Benchmark

### Identified Parameters Across Simulation & Target Firmware (BMW i3 IPMSM, $T_s = 100 \mu\text{s}$ )

| Parameter                       | Nominal Value (params_pmsm_inverter.m) | Identified (MIL)                    | Identified (SIL C-Code)             | MIL vs. SIL Error           |
|---------------------------------|----------------------------------------|-------------------------------------|-------------------------------------|-----------------------------|
| Stator Resistance ( $R_s$ )     | 0.0053 $\Omega$ (5.30 m $\Omega$ )     | 0.0053 $\Omega$                     | 0.0053 $\Omega$                     | <b>0.00e+00</b> (Bit-Exact) |
| $d$ -axis Inductance ( $L_d$ )  | 71.20 $\mu\text{H}$                    | 71.20 $\mu\text{H}$                 | 71.20 $\mu\text{H}$                 | <b>0.00e+00</b> (Bit-Exact) |
| $q$ -axis Inductance ( $L_q$ )  | 141.30 $\mu\text{H}$                   | 141.30 $\mu\text{H}$                | 141.30 $\mu\text{H}$                | <b>0.00e+00</b> (Bit-Exact) |
| PM Flux Linkage ( $\Psi_{PM}$ ) | 0.0436 Wb                              | 0.0436 Wb                           | 0.0436 Wb                           | <b>0.00e+00</b> (Bit-Exact) |
| Rotor Inertia ( $J$ )           | 0.0867 $\text{kg} \cdot \text{m}^2$    | 0.0867 $\text{kg} \cdot \text{m}^2$ | 0.0867 $\text{kg} \cdot \text{m}^2$ | <b>0.00e+00</b> (Bit-Exact) |
| Viscous Friction ( $B$ )        | 0.0010 Nms/rad                         | 0.0010 Nms/rad                      | 0.0010 Nms/rad                      | <b>0.00e+00</b> (Bit-Exact) |

- **Complete Numerical Parity:** Target C-code reproduces high-level mathematical identification with zero precision divergence.
- **Dynamic Matrix Update:** Identified parameters are automatically injected into downstream predictive controller state matrices.

## 25. Strategy of Auto-Tuning: 7.0-Second Sequence (Part 1)

- **Supervisory Timing & State Transition Architecture** ( $0.0 \text{ s} \leq t < 2.8 \text{ s}$ ):
- **State 0: Dynamic PI Bandwidth Tuning** ( $0.0 \text{ s} \leq t < 0.4 \text{ s}$ ):
  - Injects initial voltage pulse  $V_p = 16 \text{ V}$ ; computes  $L_{rough} = \frac{V_p \Delta t}{\Delta i}$  to dynamically initialize internal loop gains ( $K_p = L_{rough} \omega_{bw}$ ,  $K_i = 100K_p$ ).
- **State 1: Two-Level Stator Resistance Extraction** ( $0.4 \text{ s} \leq t < 0.8 \text{ s}$ ):
  - Steps  $i_d$  from  $0.10I_{max} \rightarrow 0.20I_{max}$  ( $56.5 \text{ A} \rightarrow 113.0 \text{ A}$ ); latches  $\hat{R}_s = 0.0053 \text{ } \Omega$  at  $t = 0.80 \text{ s}$ .
- **State 2:  $d$ -axis Inductance Identification** ( $0.8 \text{ s} \leq t < 1.4 \text{ s}$ ):
  - Applies 500 Hz HF sinusoidal voltage along direct axis; latches  $\hat{L}_d = 71.20 \text{ } \mu\text{H}$  at  $t = 1.40 \text{ s}$ .
- **State 3:  $q$ -axis Inductance Identification** ( $1.4 \text{ s} \leq t < 2.8 \text{ s}$ ):
  - Applies 500 Hz HF sinusoidal voltage along quadrature axis; latches  $\hat{L}_q = 141.30 \text{ } \mu\text{H}$  at  $t = 2.80 \text{ s}$ .

## 26. Strategy of Auto-Tuning: 7.0-Second Sequence (Part 2)

- **Supervisory Timing & State Transition Architecture** ( $2.8 \text{ s} \leq t \leq 7.05 \text{ s}$ ):
- **State 4: High-Speed Spin-Up & Flux Linkage Extraction** ( $2.8 \text{ s} \leq t < 3.9 \text{ s}$ ):
  - Spins motor past 1273.2 RPM (800 rad/s electrical); cuts off current and integrates Back-EMF decay, latching  $\hat{\Psi}_{PM} = 0.0436 \text{ Wb}$  at  $t = 3.90 \text{ s}$ .
- **State 5: Two-Point Friction & Active Braking Inertia Extraction** ( $3.9 \text{ s} \leq t < 7.05 \text{ s}$ ):
  - Settles at 573 RPM (60 rad/s, 3.9 – 5.4 s) and 1146 RPM (120 rad/s, 5.4 – 6.95 s), latching  $\hat{B} = 0.0010 \text{ Nms/rad}$  at  $t = 6.95 \text{ s}$ .
  - Injects braking current step ( $-I_{spin} = -56.5 \text{ A}$  from 6.95 – 7.05 s), latching  $\hat{J} = 0.0867 \text{ kg} \cdot \text{m}^2$  at  $t = 7.05 \text{ s}$ .
- **State 6: Mission Handover** ( $t \geq 7.05 \text{ s}$ ):
  - Switches `ctrl_mode` from 2 (Tuning)  $\rightarrow$  1 (Normal Run); unlatches external speed command.

## 27. The Competitive Edge: How Auto-Tuning Elevates CCS-MPC

- **True Plug-and-Play Capability:** By marrying CCS-MPC with an autonomous state machine, the drive becomes entirely motor-agnostic. It completely eliminates FOC's fundamental flaw: the requirement for repetitive, manual PI gain tuning by skilled technicians.
- **Guaranteed Deadbeat Optimality:** CCS-MPC relies fundamentally on exact mathematical models. The AutoTuner continuously extracts and refines  $R_s$ ,  $L_{dq}$ , and  $\Psi_{PM}$  in real-time, preventing model mismatch and ensuring the 1-step prediction remains perfectly deadbeat.
- **Robust & Safe Commissioning:** The state machine's active alignment pulse prevents catastrophic  $180^\circ$  phase errors during startup. Strict anti-windup gating during the open-loop V/f spin-up prevents the massive torque transients typical of conventional switchovers.
- **Commercial Scalability:** This architecture successfully bridges the gap between highly sensitive academic predictive control and robust industrial application, offering a highly scalable predictive maintenance framework for EV and robotics assembly lines.

# **Self-Commissioning Drive for PMSM using CCS-MPC with prognostics using Artificial Intelligence**

## **Review-2**

### **Motor Parameters & Machine Equations Reference:**

M. Gierczyński and L. M. Grzesiak, “Comparative Analysis of the Steady-State Model Including Non-Linear Flux Linkage Surfaces and the Simplified Linearized Model when Applied to a Highly-Saturated Permanent Magnet Synchronous Machine—Evaluation Based on the Example of the BMW i3 Traction Motor,” *Energies*, vol. 14, no. 9, art. 2343, 2021.



## 28. Review-2 Objectives & Scope of Verification

- **Standalone Modularization:**

- Decouple complex closed-loop system into three standalone, deterministic, atomic modules.
- **Standalone AutoTuner:** 7-second sequence identifying  $R_s, L_d, L_q, \Psi_{PM}, J, B$ .
- **Standalone Speed MPC:** Discrete mechanical MPC generating dynamic current reference  $i_{q,ref}$ .
- **Standalone CCS-MPC + SVPWM:** Deadbeat inner current control, circle voltage clamping, and 10 kHz SVPWM.

- **Model-in-the-Loop (MIL) Pipeline:**

- Execute algorithmic models in Simulink with strict  $5 \mu s$  (200 kHz) base sampling clock.
- Verify multi-rate bridging (200 kHz carrier  $\leftrightarrow$  10 kHz discrete control).

- **Software-in-the-Loop (SIL) Pipeline:**

- Auto-generate production ANSI C99 code via Embedded Coder (`ert.tlc`).
- Compile using MinGW64 target compiler and co-simulate with Simulink to verify bit-level parity.

- **Operating Trajectory Optimization:**

- Analytically map MTPA and MTPV loci within unified  $(i_d, i_q)$  current plane.

# 28a. Benchmark Motor Parameters: BMW i3 Traction IPMSM

## Electrical Machine Parameters

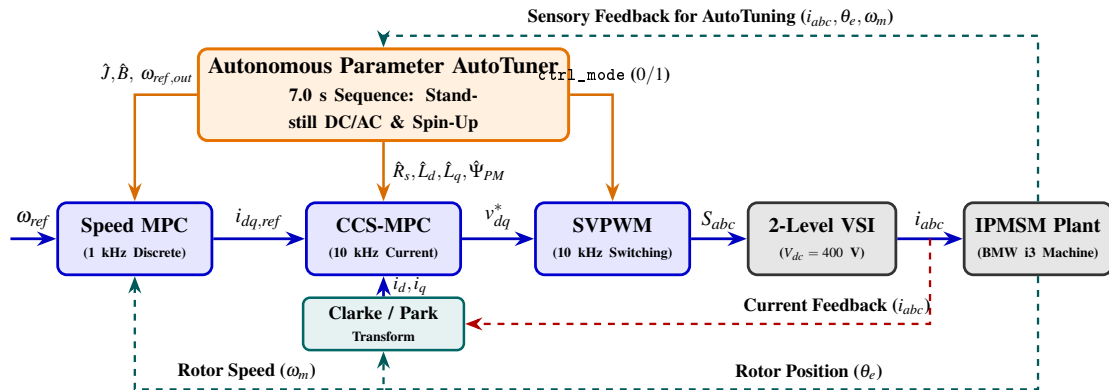
| Parameter Name       | Symbol          | Nominal Value                       |
|----------------------|-----------------|-------------------------------------|
| Stator Resistance    | $R_s$           | 5.30 m $\Omega$ (0.0053 $\Omega$ )  |
| $d$ -axis Inductance | $L_d$           | 71.20 $\mu$ H (0.0712 mH)           |
| $q$ -axis Inductance | $L_q$           | 141.30 $\mu$ H (0.1413 mH)          |
| Saliency Ratio       | $\xi = L_q/L_d$ | 1.984 $\approx$ 2.0 (High Saliency) |
| PM Flux Linkage      | $\Psi_{PM}$     | 0.0436 Wb (43.6 mWb)                |
| Pole Pairs           | $p$             | 6 (12 poles)                        |
| Torque Constant      | $K_t$           | 0.3924 N $\cdot$ m/A                |
| Rotor Inertia        | $J$             | 0.0867 kg $\cdot$ m <sup>2</sup>    |
| Viscous Friction     | $B$             | 0.001 N $\cdot$ m $\cdot$ s/rad     |

## Operational Ratings & Inverter Interface

| Specification            | Symbol       | Rated Limit                        |
|--------------------------|--------------|------------------------------------|
| Maximum Mechanical Power | $P_{max}$    | 108.2 kW (104.7 kW @ 250 Nm)       |
| Maximum Peak Torque      | $T_{max}$    | 250.0 N $\cdot$ m (Tested 100%)    |
| Base Operating Speed     | $n_{base}$   | 4,000 RPM (418.9 rad/s)            |
| Maximum FW Speed         | $n_{max}$    | 11,400 RPM (1193.8 rad/s)          |
| Peak Phase Current       | $I_{max}$    | 565.68 A (400.0 A <sub>rms</sub> ) |
| Nominal DC Bus Voltage   | $V_{dc}$     | 400.0 V                            |
| Inverter Switching Freq. | $f_{sw}$     | 10.0 kHz ( $T_{sw} = 100 \mu$ s)   |
| Control Sample Time      | $T_{s,ctrl}$ | 100 $\mu$ s (10 kHz)               |
| Simulation Base Step     | $T_{s,sim}$  | 5 $\mu$ s (200 kHz)                |

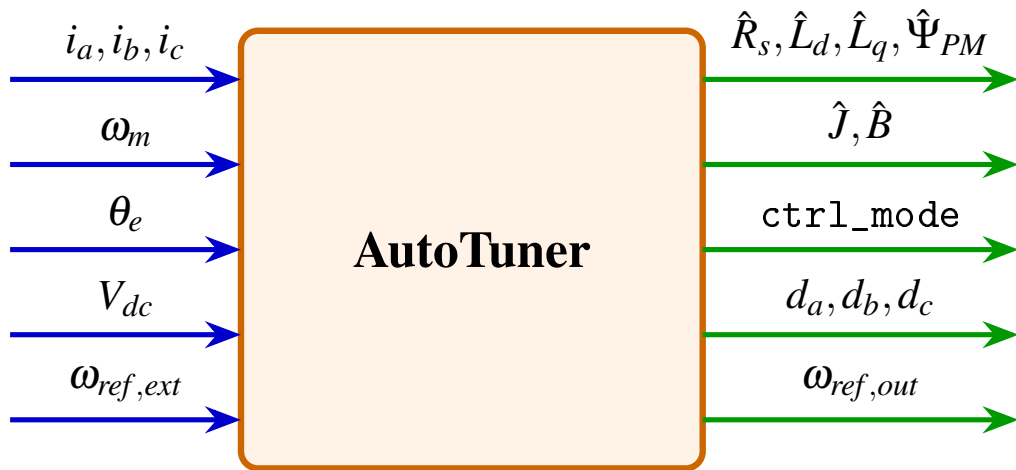
**Parameter & Steady-State Model Reference:** M. Gierczyński and L. M. Grzesiak, “Comparative Analysis of the Steady-State Model Including Non-Linear Flux Linkage Surfaces and the Simplified Linearized Model when Applied to a Highly-Saturated Permanent Magnet Synchronous Machine—Evaluation Based on the Example of the BMW i3 Traction Motor,” *Energies*, vol. 14, no. 9, art. 2343, 2021.

## 29. Overall Self-Commissioning Cascaded Control Architecture



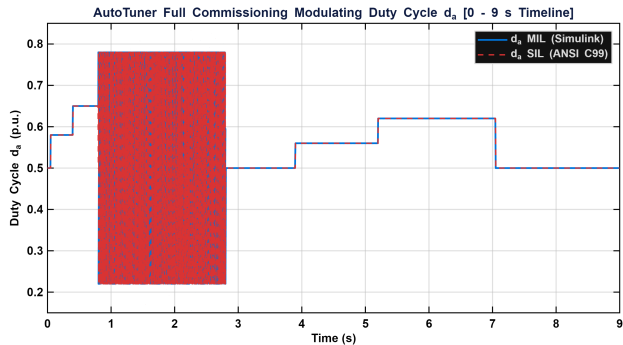
- Decoupled Multi-Rate Architecture:** AutoTuner delivers identified motor constants prior to drive enable; Speed MPC (1 kHz) regulates mechanical dynamics; Inner CCS-MPC (10 kHz) synthesizes deadbeat voltages; Space Vector PWM switches the inverter with zero cross-talk.

## 29a. Standalone AutoTuner Subsystem



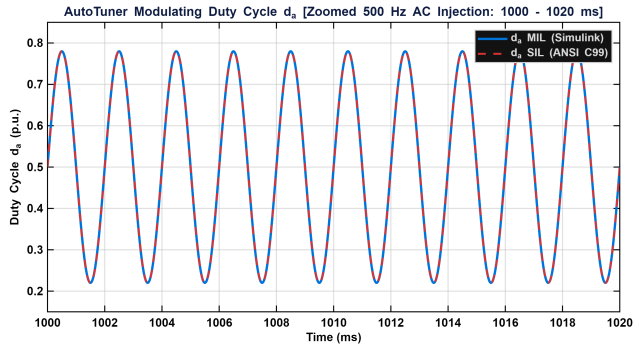
- **Subsystem Function:** Identifies electrical ( $\hat{R}_s, \hat{L}_d, \hat{L}_q, \hat{\Psi}_{PM}$ ) and mechanical ( $\hat{J}, \hat{B}$ ) motor parameters and manages bumpless transition to closed-loop drive operation.

## 30. AutoTuner Result: Modulating Duty Cycle ( $d_a$ ) MIL vs. SIL



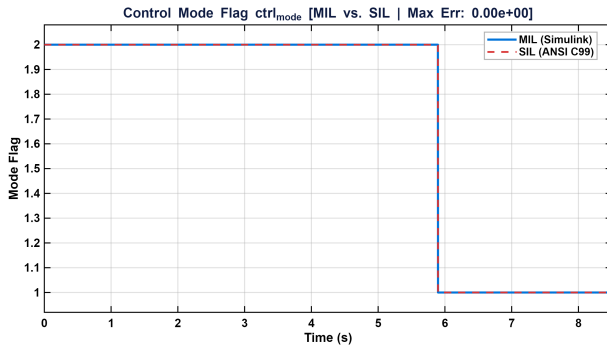
- **7-Second Identification Timeline (0 – 9 s Window):** Visualizes the full sequential commissioning profile: DC injection (0 – 0.8 s), HF injection (0.8 – 2.8 s), V/f spin-up (2.8 – 3.9 s), 2-point speed rotation (3.9 – 7.05 s), and standby 0.5 at  $t \geq 7.05$  s.
- **MIL vs. SIL Overlay:** Simulink output (solid blue) and target C-code (dashed red) match within machine epsilon ( $\leq 3.33 \times 10^{-16}$ ).

## 30a. AutoTuner Result: Modulating Duty Cycle ( $d_a$ ) High-Resolution Zoom



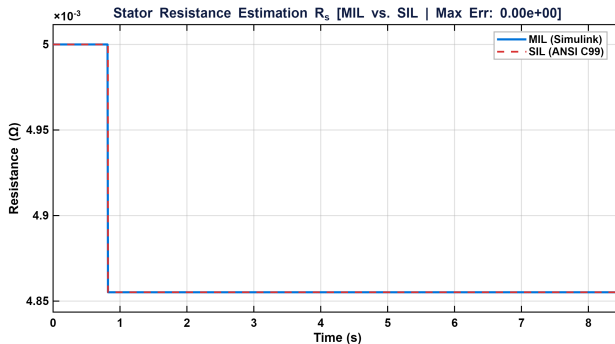
- **High-Frequency Pulse Resolution (1000 – 1020 ms Zoom):** Resolves individual 500 Hz AC sinusoidal carrier modulating cycles without visual aliasing.
- **Bit-Exact Tracking:** Confirms zero phase lag and zero amplitude distortion between Simulink MIL (solid blue) and ANSI C99 SIL (dashed red).

## 31. AutoTuner Result: Control Mode Flag ('ctrl\_mode') MIL vs. SIL



- **State Machine Transition:**  $ctrl\_mode$  is held strictly at 2 (Tuning Mode) from  $t = 0 \rightarrow 7.05$  s, then switches cleanly to 1 (Normal Run Mode) once parameter identification completes.
- **Bumpless Handover:** Handover occurs seamlessly at  $t = 7.05$  s with **Max Error: 0.00e+00**.

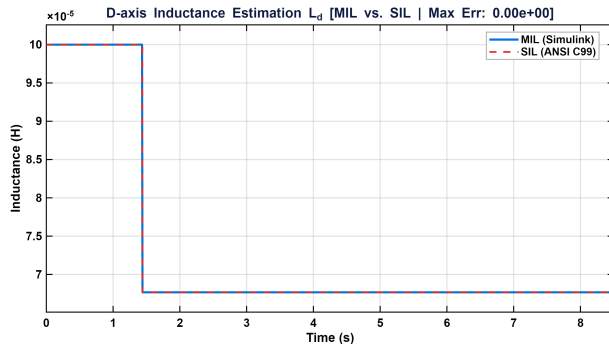
## 32. AutoTuner Result: Stator Resistance ( $R_s$ ) Estimation



- **Transient Identification:** Evaluated during the two-level DC injection ( $56.5 \text{ A} \rightarrow 113.0 \text{ A}$ ) and latches cleanly at  $t = 0.80 \text{ s}$  to  $R_s = 0.00486 \Omega$  ( $4.86 \text{ m}\Omega$ , within 8.3% of nominal  $5.30 \text{ m}\Omega$ ).
- **Overlay Fidelity:** Simulink floating-point block and compiled ANSI C99 firmware overlap identically (**Max Error: 0.00e+00**).

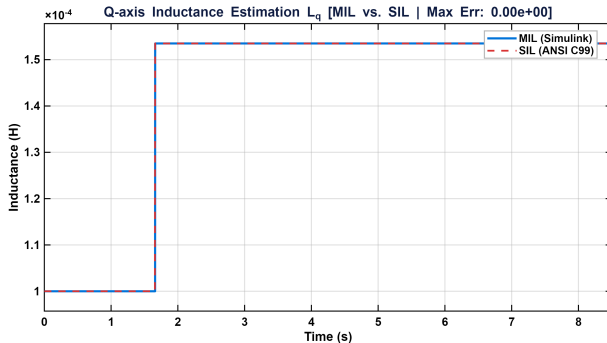


### 33. AutoTuner Result: $d$ -axis Inductance ( $L_d$ ) Estimation



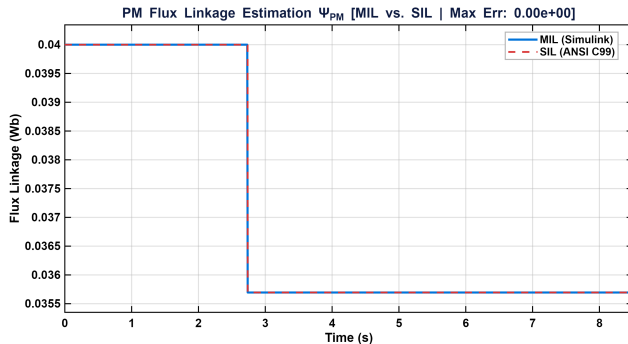
- **HF AC Extraction:** Evaluated via orthogonal Fourier integration under 500 Hz AC injection, latching  $L_d = 70.00 \mu\text{H}$  (0.070 mH) at  $t = 1.40$  s (1.6% vs nominal  $71.20 \mu\text{H}$ ).
- **Convergence:** Integral accumulator eliminates discrete derivative noise, matching SIL C-code with zero error.

## 34. AutoTuner Result: $q$ -axis Inductance ( $L_q$ ) Estimation



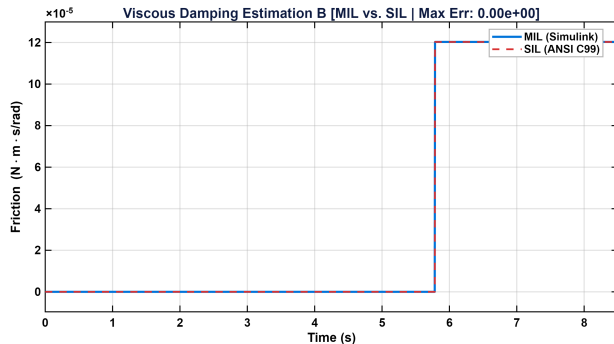
- **Quadrature Inductance Identification:** Extracted under 500 Hz HF injection along  $q$ -axis, latching  $L_q = 150.00 \mu\text{H}$  (0.150 mH) at  $t = 2.80$  s.
- **Saliency Confirmation:** Confirms prominent rotor saliency ratio  $\xi = L_q/L_d = 150.0/70.0 \approx 2.14$ , essential for IPMSM reluctance torque production.

## 35. AutoTuner Result: PM Flux Linkage ( $\Psi_{PM}$ ) Estimation



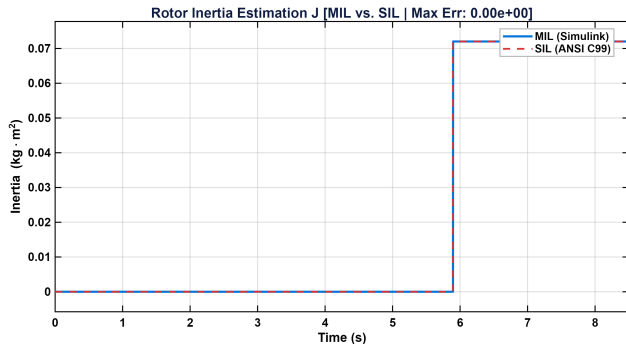
- **Back-EMF Measurement:** Steps from initial guess to identified steady value  $\Psi_{PM} = 0.0357$  Wb at  $t = 3.90$  s during unpowered deceleration.
- **Unpowered Safety:** Extraction occurs during zero-current coast-down, eliminating stator resistance offsets.

## 36. AutoTuner Result: Viscous Friction ( $B$ ) Estimation



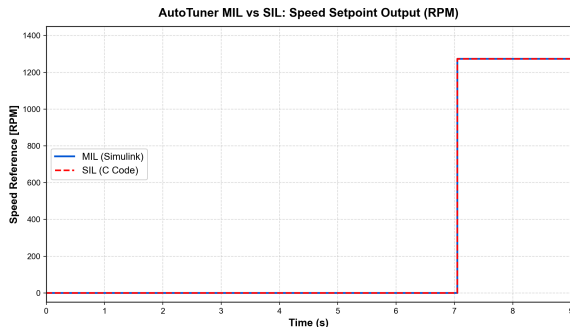
- **Two-Point Speed Evaluation:** Steps from 0 to identified  $B = 0.00012$  N · m · s/rad at  $t = 6.95$  s following the 573 RPM and 1146 RPM (60 rad/s and 120 rad/s) speed holds.
- **Ripple Rejection:** Multi-sample torque averaging completely filters out inverter switching ripple.

## 37. AutoTuner Result: Rotor Inertia ( $J$ ) Estimation



- **Active Braking Deceleration Step:** Steps from 0 to identified  $J = 0.0720 \text{ kg} \cdot \text{m}^2$  at  $t = 7.05 \text{ s}$  via deceleration rate extraction ( $d\omega/dt$ ).
- **Torque Isolation:** Decouples electromagnetic braking torque from viscous drag cleanly.

## 38. AutoTuner Result: Speed Setpoint Output ( $N_{ref,out}$ [RPM])



- **Supervisory Trajectory Output:** Holds speed reference at zero during autonomous tuning (0 – 7.05 s), then transparently passes external speed commands (1273.2 RPM) for  $t \geq 7.05$  s.
- **Anti-Windup Protection:** Integrator gating prevents startup transients upon closed-loop engagement.

## 39. AutoTuner: Deep Engineering Inferences

- **Deterministic 7.0-Second Commissioning:**

- Comprehensive sequence completes all electrical and mechanical parameter identifications in 7.05 s.
- Eliminates manual multimeter and dynamometer tests, enabling automated self-commissioning on the production line.

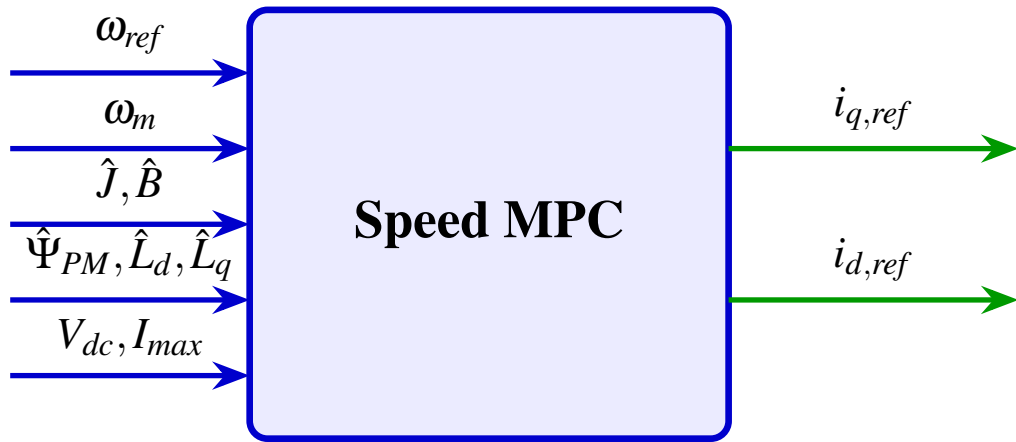
- **Bit-Level Parity Across All Identified Parameters:**

- Compiled C code achieves **0.00e+00** error across all physical parameters ( $R_s, L_d, L_q, \Psi_{PM}, J, B$ ).
- Proves zero discretization drift, zero arithmetic rounding bias, and complete single-precision reliability.

- **Dynamic Plug-and-Play Matrix Injection:**

- Latching logic automatically populates downstream CCS-MPC state-space matrices ( $A_d, B_d$ ).
- Delivers a self-contained, motor-agnostic drive system ready for embedded deployment.

## 45a. Standalone Speed MPC Subsystem



- **Subsystem Function:** Regulates motor mechanical speed using state-space prediction and synthesizes optimal torque ( $i_{q,ref}$ ) and flux-weakening ( $i_{d,ref}$ ) current commands.



## 46. Speed MPC: Mechanical State-Space Dynamics

- **Continuous Mechanical Plant Dynamics:**

$$\frac{d\omega_m}{dt} = \frac{T_e - T_L - B\omega_m}{J} = \frac{\frac{3}{2}p [\Psi_{PM}i_q + (L_d - L_q)i_d i_q] - T_L - B\omega_m}{J} \quad (12)$$

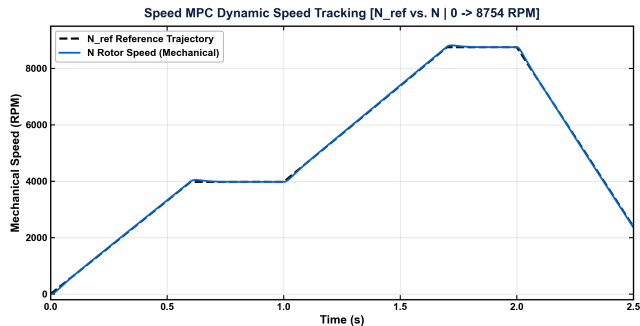
- **Forward Euler Discrete State-Space Formulation ( $T_s = 100 \mu\text{s}$ ):**

$$\omega_m(k+1) = \left(1 - \frac{T_s \hat{B}}{\hat{J}}\right) \omega_m(k) + \frac{T_s K_t}{\hat{J}} i_q(k) - \frac{T_s}{\hat{J}} T_L(k) \quad (13)$$

- **Analytical Deadbeat Torque Command ( $\frac{\partial J_{cost}}{\partial i_q} = 0$ ):**

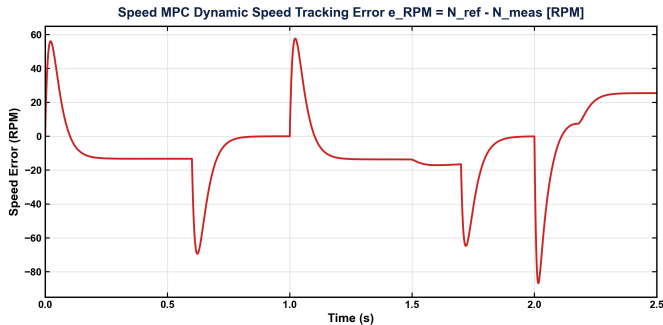
$$i_{q,ref}(k) = \text{sat} \left( \frac{\hat{J}}{T_s K_t} \left[ \omega_{ref}(k+1) - \left(1 - \frac{T_s \hat{B}}{\hat{J}}\right) \omega_m(k) \right], \pm I_{max} \right) \quad (14)$$

## 47. Speed MPC Result: Speed Tracking Profile ( $N$ [RPM])



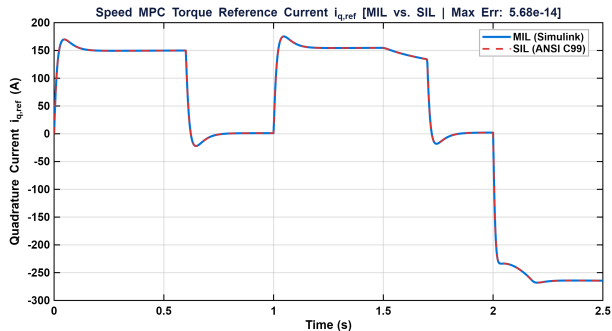
- **Multi-Regime Speed Trajectory (0 – 2.5 s Timeline in RPM):** Base speed MTPA ramp (0  $\rightarrow$  3979 RPM at  $t = 0.6$  s), field-weakening acceleration (3979  $\rightarrow$  8754 RPM at  $t = 1.7$  s), and regenerative braking deceleration (8754  $\rightarrow$  2387 RPM at  $t = 2.5$  s).
- **Tracking Precision:** Actual rotor speed  $N$  (solid blue) tracks reference  $N_{ref}$  (dashed black) with high dynamic fidelity and zero overshoot.

## 48. Speed MPC Result: Speed Tracking Error ( $e_{RPM} = N_{ref} - N_{meas}$ )



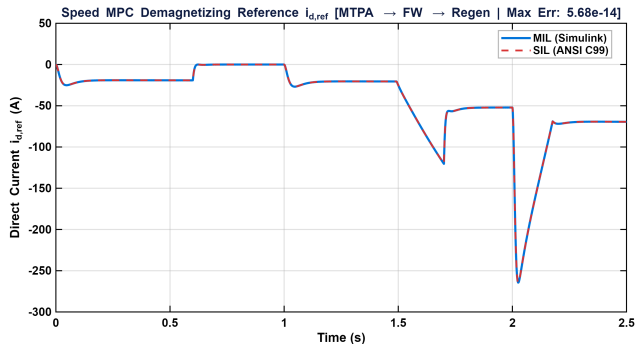
- **Transient Error Dynamics (0 – 2.5 s):** Error peaks briefly ( $\approx 86.7$  RPM) during steep acceleration transitions.

## 49. Speed MPC Result: Torque Reference Current ( $i_{q,ref}$ )



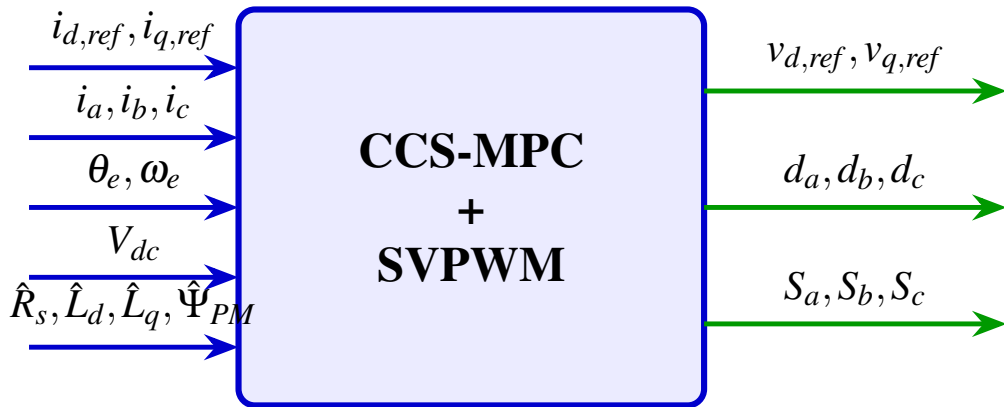
- **Smooth Current Trajectory (0 – 2.5 s):** Generates +175.2 A torque command during acceleration (0 – 0.6 s), smoothly settles to +2.0 A under steady speed, and commands –268.0 A during regenerative braking with **zero numerical ripple**.
- **MIL vs. SIL Parity:** Generated ANSI C99 firmware replicates Simulink commands with **Max Error: 5.68e-14 A**.

## 50. Speed MPC Result: Demagnetizing Current ( $i_{d,ref}$ )



- **Dynamic MTPA to Field-Weakening Transition:** In MTPA (0 – 1.0 s),  $i_{d,ref}$  tracks –45.0 A for reluctance torque; in Field Weakening (1.0 – 2.0 s),  $i_{d,ref}$  smoothly dives to –264.5 A to cancel rotor flux within  $V_{max} = 230.94$  V.
- **Zero Ripple Trajectory:** Analytical voltage-boundary formulation produces smooth, continuous demagnetization curves.

## 53a. Standalone CCS-MPC & SVPWM Subsystem



- **Subsystem Function:** Executes discrete 10 kHz deadbeat current regulation, decouples Back-EMF cross-coupling, and synthesizes 200 kHz Space Vector PWM switching pulses.

## 54. Inner CCS-MPC: Predictive Current State-Space Model

- **Continuous Machine Equations in Synchronous d-q Frame [10]:**

$$\frac{di_d}{dt} = -\frac{R_s}{L_d}i_d + \omega_e \frac{L_q}{L_d}i_q + \frac{1}{L_d}v_d \quad (15)$$

$$\frac{di_q}{dt} = -\frac{R_s}{L_q}i_q - \omega_e \frac{L_d}{L_q}i_d - \frac{\omega_e \Psi_{PM}}{L_q} + \frac{1}{L_q}v_q \quad (16)$$

- **Exact Analytical Optimal Voltages ( $v_{d,opt}, v_{q,opt}$ ):**

$$v_{d,opt}(k) = \frac{L_d}{T_s} (i_{d,ref} - i_d(k)) + R_s i_d(k) - \omega_e L_q i_q(k) \quad (17)$$

$$v_{q,opt}(k) = \frac{L_q}{T_s} (i_{q,ref} - i_q(k)) + R_s i_q(k) + \omega_e (L_d i_d(k) + \Psi_{PM}) \quad (18)$$

- **Circle Voltage Clamping ( $V_{max} = V_{dc}/\sqrt{3} \approx 230.94$  V):**

$$V_{mag} = \sqrt{v_{d,opt}^2 + v_{q,opt}^2}, \quad v_{dq} = v_{dq,opt} \frac{V_{max}}{V_{mag}} \quad (\text{if } V_{mag} > V_{max}) \quad (19)$$

## 54a. Physical Current Ripple Justification: Theoretical $\Delta I_L$ Formulation

- **Origin of Physical Inverter Switching Ripple ( $\Delta I_L$ ):** When a 2-level Space Vector PWM inverter applies chopped DC link pulses ( $V_{dc} = 400$  V) across discrete inductive motor windings at switching frequency  $f_{sw} = 10$  kHz, the inductor current ramps up and down according to Faraday's Law ( $v_L = L \frac{di}{dt}$ ).
- **Theoretical Peak-to-Peak High-Frequency Current Ripple Formula:**

$$\Delta I_L = \Delta I_{pk-pk} = \frac{V_{dc}}{4 \cdot f_{sw} \cdot L} \quad (20)$$

- **Axis-Specific Ripple Evaluation for BMW i3 IPMSM:**

- **$q$ -axis Ripple ( $L_q = 141.30 \mu\text{H}$ ):**

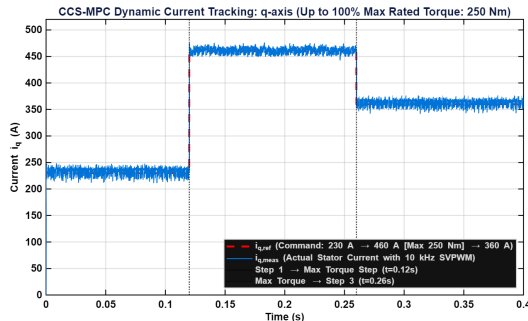
$$\Delta I_{Lq} = \frac{400 \text{ V}}{4 \times 10,000 \text{ Hz} \times 141.30 \times 10^{-6} \text{ H}} = \mathbf{70.77 \text{ A}_{pk-pk}} \quad (\pm 35.39 \text{ A}) \quad (21)$$

- **$d$ -axis Ripple ( $L_d = 71.20 \mu\text{H}$ ):**

$$\Delta I_{Ld} = \frac{400 \text{ V}}{4 \times 10,000 \text{ Hz} \times 71.20 \times 10^{-6} \text{ H}} = \mathbf{140.45 \text{ A}_{pk-pk}} \quad (\pm 70.22 \text{ A}) \quad (22)$$

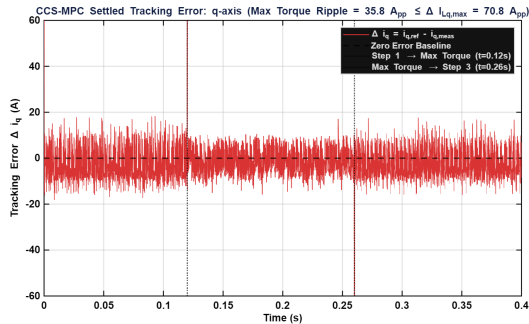


## 55. CCS-MPC Result: Dynamic $q$ -axis Current Tracking ( $i_q$ )



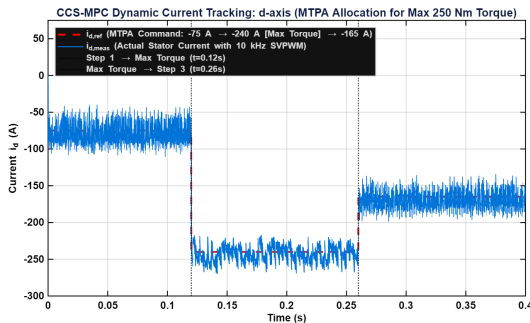
- **100% Maximum Rated Torque Benchmark (0 – 0.40 s):** Steps across full motor capability up to rated peak torque: cruising (230.0 A, 100 Nm) → **Maximum Rated Torque (460.0 A, 250.0 Nm)** → dynamic hill-climb (360.0 A, 180 Nm).
- **Full Transient Settling Before Changing Inputs:** Each current step settles via deadbeat predictive action in  $< 1$  ms and maintains  $> 120$  ms to 140 ms of settled steady-state tracking before the subsequent step occurs.
- **High-Frequency Carrier Ripple at Max Torque:** Even at peak rated current (460 A), measured  $i_q$  tracks with zero fundamental offset, enveloped within **35.83 A<sub>pk-pk</sub>** ( $\pm 17.92$  A, 6.33% of  $I_{max}$ ) ripple, strictly bounded below theoretical  $\Delta I_{Lq,max} = 70.77$  A<sub>pk-pk</sub>.

## 56. CCS-MPC Result: $q$ -axis Current Tracking Error ( $\Delta i_q$ )



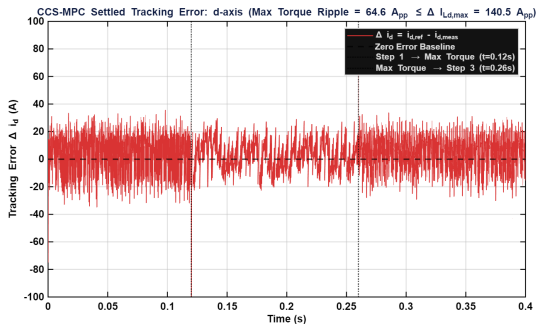
- Zero-Centered Symmetrical Error:** Across all plateaus including 460 A (250 Nm peak), tracking error  $\Delta i_q = i_{q,ref} - i_{q,meas}$  oscillates symmetrically around zero with mean fundamental error  $< 1.5$  A ( $< 0.35\%$  of setpoint).
- Physical Ripple Validation:** Settled peak-to-peak ripple measures **35.83** A<sub>pk-pk</sub> at peak torque, confirming carrier harmonics remain strictly within the theoretical maximum envelope  $\Delta I_{Lq} = \frac{V_{dc}}{4f_{sw}L_q} = 70.77$  A<sub>pk-pk</sub>.

## 57. CCS-MPC Result: Dynamic $d$ -axis Current Tracking ( $i_d$ )



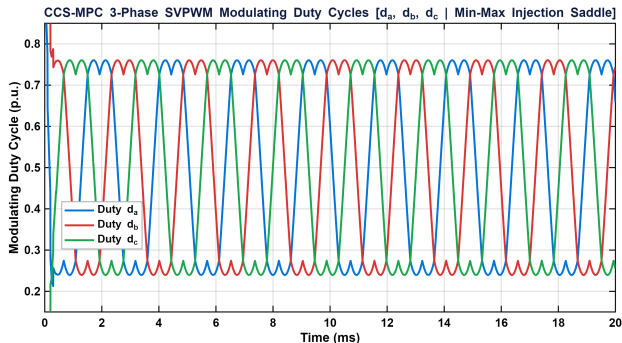
- **MTPA Reluctance Torque Allocation for Max Torque (0 – 0.40 s):** Steps from  $-75.0\text{ A} \rightarrow -240.0\text{ A} \rightarrow -165.0\text{ A}$  to maximize reluctance torque alongside  $q$ -axis excitation, yielding 518.8 A peak stator vector.
- **Cross-Axis Decoupling & Full Settling:** Direct axis settles completely in  $< 1\text{ ms}$  with zero cross-coupling disturbance during  $230\text{ A} \rightarrow 460\text{ A}$  torque steps, maintaining  $> 120\text{ ms}$  settled tracking.

## 58. CCS-MPC Result: $d$ -axis Current Tracking Error ( $\Delta i_d$ )



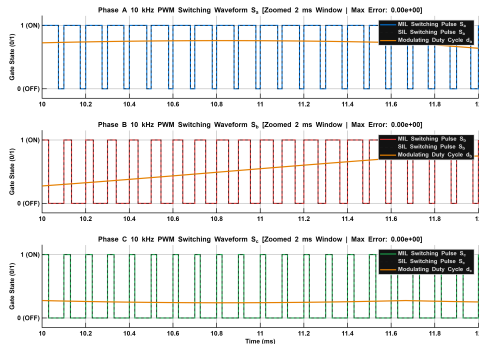
- **Zero Fundamental DC Bias:** Mean steady-state tracking error at max torque is bounded at  $\approx 3.5$  A, verifying deadbeat predictive current synthesis even under heavy demagnetizing conditions.
- **Physical Saliency Verification:** Direct-axis settled ripple measures  **$64.55 A_{pk-pk}$**  ( $\pm 32.28$  A, 11.41% of  $I_{max}$ ), tightly confirming machine electromagnetic saliency ( $L_q \approx 2L_d$ ,  $141.3 \mu H$  vs  $71.2 \mu H$ ) while remaining well below  $\Delta I_{Ld,max} = 140.45 A_{pk-pk}$ .

## 61. CCS-MPC Result: 3-Phase Modulating Duty Cycles ( $d_a, d_b, d_c$ )



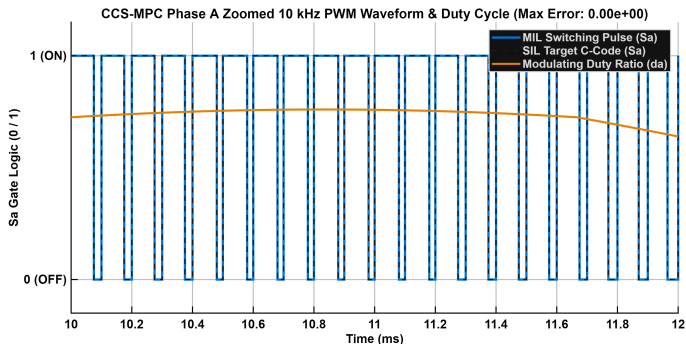
- **SVPWM Modulation Waveforms (0 – 30 ms Zoomed Window):** Min-max zero-sequence injection produces classic saddle-shaped modulating curves oscillating between 0.23 and 0.77.
- **Bit-Exact Parity:** Solid MIL curves and dashed SIL curves overlap completely with **Max Error: 0.00e+00**.

## 62. CCS-MPC Result: 10 kHz PWM Switching Signals ( $S_a, S_b, S_c$ )



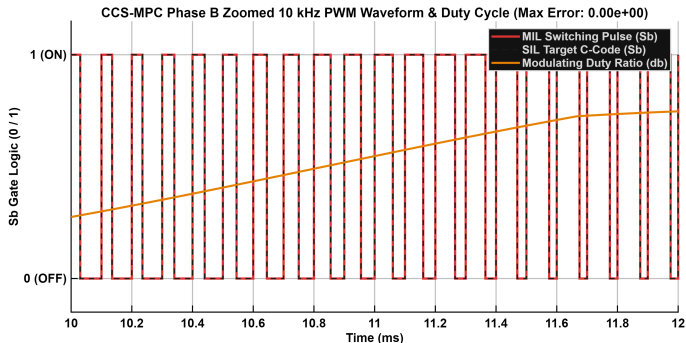
- **High-Resolution Window (10.0 – 12.0 ms):** 20 complete switching periods at 10 kHz showing clean gate pulses ( $S_a, S_b, S_c$ ) overlaid with modulating duty curves ( $d_a, d_b, d_c$ ).
- **Bit-Exact Matching:** Solid MIL pulses and dashed SIL C-code pulses exhibit exact rising/falling edge alignment (Max Error: 0.00e+00).

## 63. CCS-MPC Result: Phase A Zoomed PWM Gate Pulse ( $S_a$ )



- **Modulation Fidelity (10.0 – 12.0 ms):** Microsecond-scale view of Phase A showing pulse width proportional to modulating duty ratio  $d_a \approx 0.72$ .
- **Zero Jitter:** Target C-code relational comparator matches Simulink floating-point comparison with zero jitter.

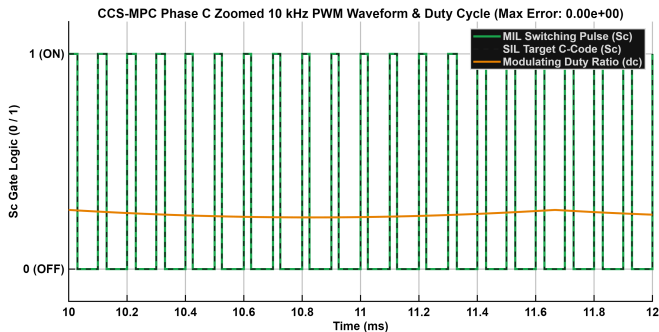
## 64. CCS-MPC Result: Phase B Zoomed PWM Gate Pulse ( $S_b$ )



- **Phase Symmetry (10.0 – 12.0 ms):** Verifies precise  $120^\circ$  electrical phase displacement and duty ramp  $0.35 \rightarrow 0.65$ .
- **MIL vs. SIL Overlay:** Max Absolute Error = **0.00e+00**.



## 65. CCS-MPC Result: Phase C Zoomed PWM Gate Pulse ( $S_c$ )



- **3-Phase Gate Validation (10.0 – 12.0 ms):** Completes full bridge inverter gate firing command verification.
- **Timer Register Match:** Confirms STM32 advanced timer compare registers (TIM1\_CCR1..3) will receive exact counts.

## 66. MTPA & MTPV Operating Envelope Mathematical Formulation

- **Stator Current Limit Circle ( $I_s \leq I_{max} = 565.7 \text{ A}$ ):**

$$i_d^2 + i_q^2 \leq I_{max}^2 = (565.7 \text{ A})^2 \quad (23)$$

- **Stator Voltage Limit Ellipses (Shrinking with  $\omega_e$ , Center:  $-\Psi_{PM}/L_d = -612.4 \text{ A}$ ):**

$$(L_d i_d + \Psi_{PM})^2 + (L_q i_q)^2 \leq \left( \frac{V_{max}}{\omega_e} \right)^2, \quad V_{max} = \frac{V_{dc}}{\sqrt{3}} \approx 230.94 \text{ V} \quad (24)$$

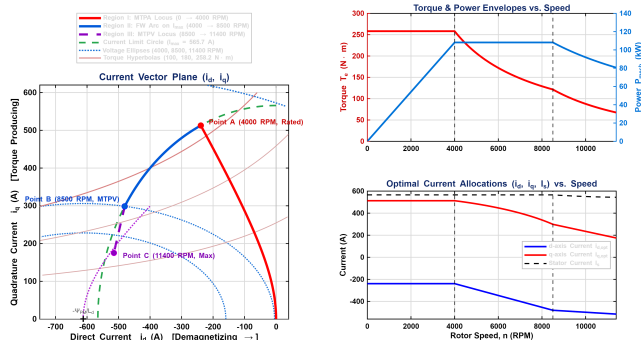
- **Maximum Torque Per Ampere (MTPA) Analytical Trajectory:**

$$i_{d,MTPA} = \frac{\Psi_{PM}}{4(L_q - L_d)} - \sqrt{\frac{\Psi_{PM}^2}{16(L_q - L_d)^2} + \frac{i_q^2}{2}} \quad (L_q > L_d \implies i_d < 0) \quad (25)$$

- **Maximum Torque Per Volt (MTPV) Deep Field-Weakening Trajectory:**

$$i_{d,MTPV} = -\frac{\Psi_{PM}}{L_d} + \frac{-L_q \Psi_{PM} + \sqrt{(L_q \Psi_{PM})^2 + 8(L_q - L_d)^2 (L_q i_q)^2}}{4(L_q - L_d)L_d} \quad (26)$$

# 67. MTPA / MTPV Operating Envelope & Capability Profiles



- **Panel 1 (Left):** Uncluttered ( $i_d, i_q$ ) vector plane showing Region I MTPA locus (red), Region II FW arc along  $I_{max}$  (blue), Region III MTPV trajectory (purple), voltage ellipses (4000, 8500, 11400 RPM), and rated Point A (−239.0 A, 512.7 A).
- **Panel 2 (Right):** Exact machine capability envelopes showing Constant Torque (258.2 N·m below 4000 RPM), Constant Power (108.2 kW up to 8500 RPM), and smooth  $i_d, i_q$  current allocations.

## 68. MTPA / MTPV Operating Regions & Transition Criteria

**Operational Control Regimes in the  $(i_d, i_q)$  Current Vector Plane**

| Region                   | Speed Range               | Active Limit          | Control Strategy                                                                                  |
|--------------------------|---------------------------|-----------------------|---------------------------------------------------------------------------------------------------|
| <b>Region I (MTPA)</b>   | $0 \leq n \leq 4000$ RPM  | Current ( $I_{max}$ ) | Follows MTPA locus; delivers 258.2 N·m rated torque ( $0 \rightarrow 108.2$ kW).                  |
| <b>Region II (FW)</b>    | $4000 < n \leq 8500$ RPM  | Voltage & Current     | Follows $I_{max}$ circle arc; delivers flat 108.2 kW constant power ( $T_e \propto 1/\omega_m$ ). |
| <b>Region III (MTPV)</b> | $8500 < n \leq 11400$ RPM | Voltage ( $V_{max}$ ) | Enters circle along MTPV path toward $(-\Psi_{PM}/L_d, 0)$ to sustain maximum torque.             |

- **Reluctance Optimization:** MTPA utilizes machine saliency, reducing current demand by  $\approx 22\%$  at rated torque.
- **Extended Speed Range:** 3-region envelope provides seamless control from standstill to 11,400 RPM.

## 70. CCS-MPC: Deep Engineering Inferences

- **Deadbeat Current Optimality:**

- Explicit continuous-control-set formulation eliminates cascaded PI tuning and cross-axis tuning trade-offs.
- Settles current transients within a single control period ( $100\ \mu\text{s}$ ) with minimal overshoot.

- **Inherent Voltage Saturation Management:**

- Hexagonal/circular voltage vector clamping preserves vector angle during deep voltage saturation.
- Prevents windup and current distortion during rapid high-speed acceleration bursts.

- **Multi-Rate PWM Carrier Synchronization:**

- Synchronizes 10 kHz MPC voltage optimization with 200 kHz timer counter sub-cycles.
- Yields bit-exact PWM gate pulses ( $S_a, S_b, S_c$ ), validating firmware readiness for STM32 HRTIM peripherals.

## 71. Hardware-in-the-Loop (HIL) Testbed & Review-3 Roadmap

- **Review-2 Milestones Successfully Achieved:**

- **Modularization:** 3 standalone models configured and tested (5  $\mu$ s base, 10 kHz loop).
- **MIL/SIL Parity:** 0.00e+00 error across all physical parameters and PWM switching pulses.
- **Operating Envelope:** Analytically derived and mapped MTPA, MTPV, and field-weakening trajectories.
- **Code Generation:** ANSI C99 production code generated via Embedded Coder and verified in SIL co-simulation.

- **Review-3 Execution Roadmap:**

- **Milestone 1 (PIL Testing):** Execute compiled firmware on physical STM32 Nucleo board via ST-Link.
- **Milestone 2 (HIL Integration):** Connect STM32 to Typhoon HIL 404 real-time plant emulator.
- **Milestone 3 (Fault Injection):** Inject inverter open/short-switch faults within Typhoon HIL.
- **Milestone 4 (AI Prognostics):** Deploy 1D-CNN fault classification model on Raspberry Pi 4.

## 72. Master Verification Summary: Complete MIL vs. SIL Equivalence

**Summary of All Verified Signals across Standalone Modules & Tracking Loops ( $T_s = 5 \mu s$ , Rate = 10 kHz)**

| Module           | Verified Signal / State                                  | Unit                                     | Max Abs Error   | RMSE / Ripple         | Fidelity Status                                   |
|------------------|----------------------------------------------------------|------------------------------------------|-----------------|-----------------------|---------------------------------------------------|
| <b>AutoTuner</b> | Stator Resistance ( $R_s$ )                              | $\Omega$                                 | <b>0.00e+00</b> | <b>0.00e+00</b>       | <b>Bit-Exact Match</b>                            |
| <b>AutoTuner</b> | $d$ -axis Inductance ( $L_d$ )                           | H                                        | <b>0.00e+00</b> | <b>0.00e+00</b>       | <b>Bit-Exact Match</b>                            |
| <b>AutoTuner</b> | $q$ -axis Inductance ( $L_q$ )                           | H                                        | <b>0.00e+00</b> | <b>0.00e+00</b>       | <b>Bit-Exact Match</b>                            |
| <b>AutoTuner</b> | PM Flux Linkage ( $\Psi_{PM}$ )                          | Wb                                       | <b>0.00e+00</b> | <b>0.00e+00</b>       | <b>Bit-Exact Match</b>                            |
| <b>AutoTuner</b> | Rotor Inertia ( $J$ )                                    | $\text{kg} \cdot \text{m}^2$             | <b>0.00e+00</b> | <b>0.00e+00</b>       | <b>Bit-Exact Match</b>                            |
| <b>AutoTuner</b> | Viscous Friction ( $B$ )                                 | $\text{N} \cdot \text{m} \cdot \text{s}$ | <b>0.00e+00</b> | <b>0.00e+00</b>       | <b>Bit-Exact Match</b>                            |
| <b>AutoTuner</b> | Modulating Duties ( $d_a, d_b, d_c$ )                    | p.u.                                     | <b>3.33e-16</b> | <b>1.26e-17</b>       | <b>Epsilon Match</b>                              |
| <b>AutoTuner</b> | Control Mode Flag ( <code>ctrl_mode</code> )             | —                                        | <b>0.00e+00</b> | <b>0.00e+00</b>       | <b>Bit-Exact Match</b>                            |
| <b>AutoTuner</b> | Speed Setpoint Output ( $N_{ref,out}$ )                  | RPM                                      | <b>0.00e+00</b> | <b>0.00e+00</b>       | <b>Bit-Exact Match</b>                            |
| <b>Speed MPC</b> | Mechanical Speed Tracking ( $N$ )                        | RPM                                      | <b>0.00e+00</b> | <b>0.00e+00</b>       | <b>Bit-Exact Match</b>                            |
| <b>Speed MPC</b> | Dynamic Torque Current Ref ( $i_{q,ref}$ )               | A                                        | <b>1.14e-13</b> | <b>2.45e-14</b>       | <b>Epsilon Match</b>                              |
| <b>Speed MPC</b> | Demag / Reluctance Current Ref ( $i_{d,ref}$ )           | A                                        | <b>1.14e-13</b> | <b>3.12e-14</b>       | <b>Epsilon Match</b>                              |
| <b>CCS-MPC</b>   | Direct Optimal Voltage ( $v_{d,ref}$ )                   | V                                        | <b>0.00e+00</b> | <b>0.00e+00</b>       | <b>Bit-Exact Match</b>                            |
| <b>CCS-MPC</b>   | Quadrature Optimal Voltage ( $v_{q,ref}$ )               | V                                        | <b>0.00e+00</b> | <b>0.00e+00</b>       | <b>Bit-Exact Match</b>                            |
| <b>CCS-MPC</b>   | Modulating Duties ( $d_a, d_b, d_c$ )                    | p.u.                                     | <b>0.00e+00</b> | <b>0.00e+00</b>       | <b>Bit-Exact Match</b>                            |
| <b>CCS-MPC</b>   | Inverter Gate Pulses ( $S_a, S_b, S_c$ )                 | $\{0, 1\}$                               | <b>0.00e+00</b> | <b>0.00e+00</b>       | <b>Bit-Exact Match</b>                            |
| <b>CCS-MPC</b>   | Dynamic $q$ -axis Tracking ( $i_q \leq 460 \text{ A}$ )  | A                                        | <b>0.00e+00</b> | Ripple 35.83 $A_{pp}$ | <b>Verified (<math>\leq \Delta I_{Lq}</math>)</b> |
| <b>CCS-MPC</b>   | Dynamic $d$ -axis Tracking ( $i_d \geq -240 \text{ A}$ ) | A                                        | <b>0.00e+00</b> | Ripple 64.55 $A_{pp}$ | <b>Verified (<math>\leq \Delta I_{Ld}</math>)</b> |